



POLITECNICO
MILANO 1863

Pneumonia Detection Using Anchor-Free vs. Anchor-Based Object Detection

Reproducing and Extending Wu et al. (2024)

Numerical Analysis for Machine Learning — Politecnico di Milano

Mohamed Z. M. Mandour

mohamedzeyad.mandour@mail.polimi.it

Elena Nuttini

elena.nuttini@mail.polimi.it

June 2026

Abstract

This report reproduces and extends the method proposed by Wu et al. [1], which applies the anchor-free FCOS (Fully Convolutional One-Stage) detector to pneumonia detection on chest X-rays from the RSNA Pneumonia Detection Challenge dataset. We implement and compare four detector variants sharing the same ResNet-50 + Feature Pyramid Network (FPN) backbone: (1) FCOS with the paper’s anchor-free head trained under our modern Adam recipe, (2) the same FCOS head retrained under the paper’s original SGD+momentum recipe as an optimiser ablation, (3) RetinaNet as a one-stage anchor-based baseline, and (4) Faster R-CNN as a two-stage anchor-based baseline. Training is performed on the full RSNA dataset (26,684 images) for 40 epochs on four NVIDIA H100 80 GB SXM GPUs (one model per GPU) with BF16 mixed-precision, cosine annealing, warm-up, and Exponential Moving Average (EMA) of the weights. At evaluation time we apply horizontal-flip Test-Time Augmentation (TTA) and Gaussian Soft-NMS, and combine FCOS with RetinaNet via Weighted Box Fusion (WBF).

All detectors *exceed* the paper’s reported FCOS AP@0.5 of 28.5%: our FCOS (Adam) reaches 35.3% (+6.8), FCOS with the paper’s SGD recipe reaches 39.8% (+11.3), RetinaNet reaches 41.2% (+12.7), and Faster R-CNN reaches 42.9% (+14.4). The central empirical finding is that, under a shared backbone and a shared *Adam* recipe, the anchor-based detectors decisively outperform the anchor-free FCOS on box-level AP@0.5; switching FCOS to the paper’s SGD recipe narrows but does *not* close the gap to RetinaNet (Section 10 reports the paired bootstrap on the per-patient difference). Crucially, this is an *asymmetric* optimiser ablation — we only re-trained FCOS under SGD — so the result tightens the FCOS-vs-RetinaNet comparison without resolving the underlying paradigm-vs-optimiser confound. At the patient level, the WBF ensemble reaches F1 63.0% at the fixed threshold $\tau = 0.3$ used by the paper, which is +13.5 points above the best single model *at that arbitrary cut*; this headline is largely a calibration artefact, as Section 10 demonstrates by refitting a Youden-optimal threshold on a held-out calibration half and by training a 5-fold-CV logistic aggregator on per-patient features. Under either fair protocol the ensemble’s lead shrinks to ≤ 1 F1 point and is overtaken by RetinaNet (learnt aggregator F1 67.3%, the single best patient-level number in the study). All point estimates carry two caveats that the bootstrap does not address: they come from a single training seed, so run-to-run noise is uncovered; and they are read at each detector’s best-validation EMA checkpoint, on the same split the bootstrap CIs are computed on. We discuss both in Section 13.

Contents

1	Introduction	4
2	Related Work	6
3	Background	7
3.1	Object Detection Paradigms	7
3.2	Feature Pyramid Network (FPN)	7
3.3	Focal Loss	7
3.4	Fully Convolutional One-Stage (FCOS) Detection	7
3.5	RetinaNet: Anchor-Based Dense Prediction with Focal Loss	8
3.6	Faster R-CNN: Two-Stage Detection	8
3.7	Exponential Moving Average of Weights	8
3.8	Non-Maximum Suppression and Its Variants	9
3.9	Weighted Box Fusion for Ensembling	9
4	Dataset	10
5	Method	11
5.1	Model Architectures	11
5.2	Training Configuration	11
5.3	Evaluation Enhancements	11
5.4	Hardware and Parallelization	12
6	Training Protocol	13
6.1	Transfer Learning and Backbone Freezing	13
6.2	Learning Rate Schedule	13
6.3	Exponential Moving Average at Validation Time	13
6.4	Class Imbalance and Weighted Sampling	13
6.5	Validation Frequency and Early Stopping	13
6.6	Mixed Precision and Numerical Stability	14
7	Evaluation Metrics	15
7.1	Detection Metrics	15
7.2	Threshold-Independent Ranking Quality	15
7.3	Patient-Level Classification	15
8	Results	17
8.1	Detection Performance	17
8.2	Training Dynamics	18
8.3	Comparison Plots	19
8.4	Detection Samples	21
8.5	Patient-Level Classification	23
9	Ablation and Component Analysis	24
10	Robustness Checks	26
10.1	Patient-level Bootstrap Confidence Intervals	26
10.2	Three-way Split: Threshold on Calibration Half, F1 on Held-out Half	27
10.3	RSNA-percentile Size Buckets	27
10.4	FROC Curve	28
10.5	Calibration: Reliability Diagrams and Expected Calibration Error	28

10.6	Learnt Patient-Level Aggregator	29
10.7	Paired Bootstraps on AP@0.5 Differences	29
11	Numerical Analysis of the Training Recipe	32
11.1	EMA as a Linear Low-Pass Filter	32
11.2	Focal Loss Gradient and Effective Sample Weight	32
11.3	GIoU as a Differentiable Surrogate for IoU	33
11.4	Cosine Annealing with Warm-Up	33
11.5	Bootstrap Confidence Intervals: Monte Carlo Convergence	34
12	Discussion	35
12.1	Anchor-Free vs. Anchor-Based: A Revised View	35
12.2	What Did the Paper Get Right?	35
12.3	Clinical Interpretation of the Numbers	36
12.4	Cost and Reproducibility	36
13	Limitations and Future Work	37
13.1	Limitations	37
13.2	Future Work	37
14	Conclusion	39

1 Introduction

Pneumonia is one of the leading causes of death worldwide, particularly among children and the elderly. Accurate and timely detection from chest radiographs is critical but challenging, as pneumonia opacities can be subtle, overlapping with other pathologies, and difficult to localize precisely.

Deep learning-based object detection has shown great promise for automating pneumonia detection. Most prior approaches use *anchor-based* detectors such as Faster R-CNN [6] or RetinaNet [3], which rely on predefined anchor boxes to propose candidate regions. In contrast, Wu et al. [1] proposed using FCOS [2], an *anchor-free* detector that directly predicts bounding boxes at each spatial location of the feature map, eliminating the need for hand-designed anchor configurations.

The key advantages of the anchor-free approach include:

- No hyperparameter tuning for anchor shapes, sizes, or aspect ratios.
- More flexible localization, particularly for irregularly-shaped or variable-size lesions.
- Simpler architecture with fewer design choices.

In this project, we implement and compare three detection architectures:

1. **FCOS** — the paper’s proposed anchor-free detector,
2. **RetinaNet** — a one-stage, anchor-based detector with focal loss,
3. **Faster R-CNN** — a two-stage, anchor-based detector with RPN.

All models share a ResNet-50 backbone with Feature Pyramid Network (FPN), ensuring that performance differences arise from the detection paradigm rather than feature extraction capacity.

Contributions

- A re-implementation of Wu et al. [1] on the full RSNA dataset (we do *not* claim a bit-for-bit reproduction: the paper releases neither code nor weights, and we modernise the recipe with BF16, EMA, cosine annealing and medical-specific augmentations that the paper does not use). Our FCOS-Adam crosses the paper’s reported 28.5% AP@0.5 at epoch 12 and ends +6.8 points above it at the 40-epoch convergence; the SGD-recipe ablation reaches +11.3 points above the paper number.
- A same-backbone, same-budget comparison of three representative detection paradigms (anchor-free FCOS, one-stage anchor-based RetinaNet, two-stage anchor-based Faster R-CNN) under a shared *Adam* recipe, plus an FCOS-only SGD ablation that replicates the optimiser of Wu et al. The Adam-trained FCOS is decisively below the anchor-based detectors on AP@0.5; under the paper SGD recipe the FCOS-vs.-RetinaNet gap narrows. We have *not* re-trained RetinaNet or Faster R-CNN under SGD, so the comparison separates paradigm from optimiser only on the FCOS side.
- A Weighted Box Fusion ensemble of the two one-stage detectors (FCOS + RetinaNet) that improves patient-level F1 at the fixed $\tau = 0.3$ cut without retraining; under a fair per-detector operating point (Section 10) the ensemble matches but does *not* dominate the best single detector.
- A detailed analysis of the score-calibration mismatch between the four detectors that explains why fixed-threshold patient-level F1 differs much more than threshold-free ROC AUC — a confound that the paper does not address — together with a learnt 5-fold cross-validated aggregator that essentially closes the F1 gap.
- A public PyTorch implementation with the training recipes (cosine + warm-up schedule, EMA, medical-imaging augmentation pipeline) and all evaluation artefacts released with the report. Only the patient-level split (seed 42) is bit-deterministic; the training run itself is not, because TF32 + BF16 + asynchronous dataloader workers + nondeterministic CUDA atomics break exact reproducibility.

Outline. Section 2 surveys object detection and its use in medical imaging. Section 3 introduces the mathematical background: FPN, focal loss, EMA, (Soft-)NMS, and WBF. Section 4 describes the RSNA dataset and our preprocessing. Section 5 presents the three detector heads formally and Section 6 details the training recipe. Section 7 defines the metrics. Section 8 reports the quantitative and qualitative results, Section 9 provides an analysis of each component’s contribution, Section 10 runs seven post-hoc robustness checks on the cached predictions, and Section 11 consolidates the numerical-analysis content of the training recipe (EMA as a linear low-pass filter, focal loss gradient analysis, GIoU as a differentiable surrogate for IoU, cosine annealing with warm-up, and bootstrap Monte Carlo convergence). Section 12 discusses the findings including the perhaps surprising anchor-based win, and Section 13 covers limitations and future directions.

2 Related Work

Two-stage anchor-based detection. The modern two-stage detector lineage begins with R-CNN [8], which extracts region proposals with Selective Search and classifies each proposal with a CNN. Fast R-CNN [9] replaces per-proposal feature extraction with a shared feature map and ROI pooling. Faster R-CNN [6] learns the proposals jointly via a Region Proposal Network (RPN), becoming the canonical two-stage detector. Subsequent refinements include ROI Align (Mask R-CNN [10]) and Cascade R-CNN [11], which progressively refine proposals with increasing IoU thresholds.

One-stage anchor-based detection. One-stage detectors skip the proposal stage and predict class + box offsets directly from a dense grid of anchors. SSD [12] and YOLO [13] pioneered this design. RetinaNet [3] identified *class imbalance* as the core reason one-stage detectors underperformed their two-stage counterparts, and introduced focal loss to down-weight easy background examples; this closed the accuracy gap while retaining one-stage speed.

Anchor-free detection. Anchor-free detectors eliminate hand-crafted anchor hyperparameters (sizes, aspect ratios, per-level assignment). CornerNet [14] detects objects as pairs of corners. FCOS [2], which we adopt in this work, predicts object distances from each feature-map location, plus a *center-ness* score that down-weights predictions far from the object centre. CenterNet [15], FoveaBox [16] and ATSS [17] refined the anchor-free paradigm, with ATSS further arguing that the anchor-free / anchor-based divide is narrower than previously thought and that positive-sample assignment is the dominant factor.

Medical-image object detection. Early approaches for pneumonia and lung-lesion detection on chest X-rays used hand-crafted features coupled with an SVM or random forest classifier. Deep learning for thoracic disease localisation on chest radiographs was popularised by Wang et al. [19] with the ChestX-ray14 dataset, on which weakly supervised localisation achieved modest performance. The 2018 RSNA Pneumonia Detection Challenge [7] catalysed the use of modern detection architectures on chest X-rays, with top solutions relying on Faster R-CNN or RetinaNet with heavy ensembling. The reference paper of this work, Wu et al. [1], was among the first to apply an anchor-free FCOS pipeline to RSNA and report a favourable comparison against RetinaNet and Faster R-CNN.

Ensembling detection results. Detection ensembles must merge bounding boxes from heterogeneous models, which standard NMS does not handle well. Soft-NMS [20] replaces hard removal with a score-decay based on IoU, preserving partially overlapping detections. Weighted Box Fusion (WBF) [21] clusters boxes from multiple models and returns score-weighted averages of the box coordinates; on PASCAL VOC and COCO it consistently outperforms NMS-based ensembling. We use WBF to combine FCOS and RetinaNet predictions in Section 8.

Training recipes. Several techniques that are not specific to detection have become standard in modern recipes and make a measurable difference on our task: cosine learning-rate annealing [22], BF16 or FP16 automatic mixed precision [25], and Exponential Moving Average (EMA) of the weights [23, 24]. We adopt all three.

3 Background

3.1 Object Detection Paradigms

Anchor-Based Detection. Traditional object detectors rely on a set of predefined *anchor boxes* at each spatial location of the feature map. The network classifies each anchor as object/background and regresses offsets to refine the box. Faster R-CNN uses a Region Proposal Network (RPN) as a first stage to generate proposals, followed by ROI pooling and per-proposal classification/regression (two-stage). RetinaNet is a one-stage detector that directly classifies and regresses from anchors, using focal loss to address class imbalance.

Anchor-Free Detection. FCOS predicts, at each location (x, y) of the feature map, a classification score and four distances (l, t, r, b) from the point to the bounding box boundaries. A “center-ness” branch further down-weights predictions far from the object center. This eliminates all anchor-related hyperparameters.

3.2 Feature Pyramid Network (FPN)

All three architectures use FPN [4] on top of ResNet-50 [5] to build a multi-scale feature representation. FPN constructs a top-down pathway with lateral connections, producing feature maps at strides $\{8, 16, 32, 64, 128\}$ pixels.

3.3 Focal Loss

Both FCOS and RetinaNet employ focal loss [3] to handle the extreme foreground–background imbalance inherent in dense object detection. For a predicted probability $p \in [0, 1]$ of the ground-truth class:

$$p_t = \begin{cases} p & \text{if } y = 1 \\ 1 - p & \text{otherwise} \end{cases} \quad \text{FL}(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t), \quad (1)$$

where α_t is a class-balancing factor and $\gamma \geq 0$ is the focusing parameter. For $\gamma = 0$ focal loss reduces to weighted cross-entropy; for $\gamma = 2$ (the default, which we use) the loss on easy examples ($p_t \rightarrow 1$) is down-weighted by a factor of $\approx (1 - p_t)^2$, so that the dominant background class contributes much less to the gradient and the network focuses learning on hard foregrounds.

3.4 Fully Convolutional One-Stage (FCOS) Detection

FCOS [2] drops anchor boxes entirely. For every spatial location (x, y) on an FPN feature map, it predicts:

- a classification score $p_{x,y}$ (logits, trained with focal loss),
- four regression distances (l^*, t^*, r^*, b^*) from the point to the box edges (trained with GIoU loss),
- a *center-ness* score $c_{x,y} \in [0, 1]$ whose regression target is

$$c_{x,y}^* = \sqrt{\frac{\min(l^*, r^*)}{\max(l^*, r^*)} \cdot \frac{\min(t^*, b^*)}{\max(t^*, b^*)}}, \quad (2)$$

which is 1 at the ground-truth centre and falls off to 0 at the edges.

At inference, the per-location score used for ranking is $p_{x,y} \cdot c_{x,y}$, so boxes emitted by off-centre locations are penalised. To avoid ambiguous positive assignment when multiple ground-truth boxes cover the same location, FCOS restricts each FPN level to boxes of a certain scale range: $[0, 64, 128, 256, 512, \infty]$ in feature-map units for levels P_3 – P_7 . This multi-scale assignment is the anchor-free analogue of anchor scale priors.

The full FCOS loss is

$$\begin{aligned}\mathcal{L}_{\text{FCOS}} = & \frac{1}{N_{\text{pos}}} \sum_{x,y} \text{FL}(p_{x,y}, y_{x,y}^*) \\ & + \frac{\lambda}{N_{\text{pos}}} \sum_{x,y} \mathbb{1}_{\{y_{x,y}^* > 0\}} \text{GIoU}(t_{x,y}, t_{x,y}^*) \\ & + \frac{1}{N_{\text{pos}}} \sum_{x,y} \mathbb{1}_{\{y_{x,y}^* > 0\}} \text{BCE}(c_{x,y}, c_{x,y}^*),\end{aligned}\tag{3}$$

where N_{pos} is the number of positive locations, $\lambda = 1$ in our setting, and $t_{x,y}$ is the regressed box encoded as (l, t, r, b) .

3.5 RetinaNet: Anchor-Based Dense Prediction with Focal Loss

RetinaNet [3] places $A = 9$ anchors (3 scales $\times$ 3 aspect ratios) at each FPN location. Anchors are matched to ground-truth boxes by IoU: anchors with $\text{IoU} > 0.5$ with any ground truth are positive, those with $\text{IoU} < 0.4$ are negative, and the remainder are ignored. The classification sub-network outputs $K \cdot A$ logits per location (for K classes, $K = 1$ in our case: pneumonia), trained with focal loss. The regression sub-network outputs $4A$ offsets per location, trained with a smooth-L1 loss on the standard (t_x, t_y, t_w, t_h) parameterisation:

$$t_x = \frac{x - x_a}{w_a}, \quad t_y = \frac{y - y_a}{h_a}, \quad t_w = \log \frac{w}{w_a}, \quad t_h = \log \frac{h}{h_a}.\tag{4}$$

Unlike FCOS, RetinaNet’s positive/negative assignment is anchor-based and independent of the ground-truth centre. The v2 implementation we use [26] replaces smooth-L1 with GIoU regression and adjusts the feature extractor for slightly improved COCO accuracy.

3.6 Faster R-CNN: Two-Stage Detection

Faster R-CNN [6] splits detection into two stages sharing the backbone:

1. The **Region Proposal Network (RPN)** slides a small conv network over the FPN feature maps and, at each location, predicts an objectness score and bounding-box offsets from $A = 3$ reference anchors per level. The top- k proposals (2000 at training, 1000 at inference in our configuration) are kept after NMS.
2. The **ROI head** crops each proposal from the FPN feature maps via *ROI Align* [10], passes the result through two fully-connected layers, and emits a per-class logit and a per-class regression. Only the top-300 scored proposals are retained at test time.

Both stages use smooth-L1 regression and a 1:3 positive:negative sampling ratio in the losses. The v2 implementation [26] we use replaces smooth-L1 with GIoU and tightens the training recipe. Two-stage refinement typically yields higher recall at the cost of longer inference, which we confirm experimentally.

3.7 Exponential Moving Average of Weights

EMA [23, 24] maintains a shadow copy $\bar{\theta}_t$ of the model parameters that is updated after each optimiser step with decay μ :

$$\bar{\theta}_t = \mu \bar{\theta}_{t-1} + (1 - \mu) \theta_t.\tag{5}$$

The EMA weights are a smoothed trajectory of the SGD iterates and typically give a lower validation loss than the raw iterate, especially near convergence where the optimiser oscillates around a minimum. We use $\mu = 0.999$, so the effective window is $\sim 1/(1 - \mu) = 1000$ steps.

3.8 Non-Maximum Suppression and Its Variants

Classical greedy NMS removes a predicted box b_j if its IoU with a higher-scored surviving box b_i of the same class exceeds a threshold T_{NMS} (we use $T_{\text{NMS}} = 0.5$):

$$s_j \leftarrow \begin{cases} s_j & \text{if } \text{IoU}(b_i, b_j) < T_{\text{NMS}} \\ 0 & \text{otherwise} \end{cases}. \quad (6)$$

This *hard* suppression is brittle when two true objects legitimately overlap, a common situation in pneumonia where bilateral consolidations can overlap in the 2D projection. **Soft-NMS** [20] instead decays the score continuously:

$$s_j \leftarrow s_j \cdot \exp\left(-\frac{\text{IoU}(b_i, b_j)^2}{\sigma}\right), \quad (7)$$

with $\sigma = 0.5$ in our runs. Detections are then filtered by a minimum score. We apply Soft-NMS to all three models at evaluation; the effect is typically +0.5 to +1.5 AP.

3.9 Weighted Box Fusion for Ensembling

Given M models each producing a list of predictions per image, Weighted Box Fusion (WBF) [21] clusters boxes that match in IoU and label, and fuses each cluster C into a single box:

$$\tilde{b}_C = \frac{\sum_{(b_i, s_i) \in C} s_i b_i}{\sum_{(b_i, s_i) \in C} s_i}, \quad \tilde{s}_C = \frac{\sum_{(b_i, s_i) \in C} w_i s_i}{\sum_{m=1}^M w_m} \cdot \frac{\min(|C|, M)}{M}, \quad (8)$$

where w_m is the weight assigned to model m (uniform in our experiments). The second factor of $\tilde{s}_C$ is a *coverage* penalty that down-weights clusters missed by some of the models. Our implementation clusters greedily in descending score order with an IoU threshold of 0.55 and a pre-filter of $s_i > 0.01$.

4 Dataset

We use the **RSNA Pneumonia Detection Challenge** dataset [7], which consists of 26,684 chest X-ray images in DICOM format with bounding box annotations for pneumonia opacities. Each image is associated with a patient ID and may contain zero or more bounding boxes.

Preprocessing. DICOM images are converted to 8-bit PNG format for efficient loading. Pixel values are extracted from the DICOM metadata with proper windowing applied where available. All images are resized to 512×512 pixels.

Data Split. We perform a patient-level 80/20 train/validation split, ensuring that no patient appears in both sets. The full dataset is used for training.

Data Augmentation. We employ a medical imaging-specific augmentation pipeline using the `albumentations` library, going beyond the paper’s basic augmentations to improve generalization:

- **CLAHE** (Contrast Limited Adaptive Histogram Equalization, $p = 0.3$): enhances local contrast in X-ray images, a standard technique in medical imaging.
- **RandomBrightnessContrast** (± 0.2 , $p = 0.4$): simulates exposure and gain variation.
- **RandomGamma** (80–120, $p = 0.2$): simulates display gamma differences.
- **HorizontalFlip** ($p = 0.5$): anatomically valid since lungs are roughly symmetric. No vertical flip is used, as chest X-rays are always upright.
- **Affine** (shift 5%, scale $\pm 10\%$, rotate $\pm 10^\circ$, $p = 0.3$): simulates patient positioning variation.
- **GaussianBlur** / **GaussNoise** ($p = 0.2$, `OneOf`): simulates acquisition noise and defocus.
- **GridDistortion** / **OpticalDistortion** ($p = 0.15$, `OneOf`): simulates anatomical variation.
- **CoarseDropout** (1–4 holes, 16–32 px, $p = 0.15$): regularization via random occlusion.

All geometric transforms are applied jointly to both images and bounding boxes, with boxes smaller than 100 px^2 or less than 30% visible after augmentation automatically discarded.

5 Method

5.1 Model Architectures

All three models use the following shared configuration:

- **Backbone:** ResNet-50 pretrained on ImageNet (IMAGENET1K_V1)
- **Neck:** Feature Pyramid Network (FPN) with 5 levels
- **Input size:** 512×512 pixels
- **Classes:** 2 (background + pneumonia)
- **NMS threshold:** 0.5
- **Score threshold:** 0.05 (inference)

FCOS (Anchor-Free). The FCOS head predicts, at each FPN level, a classification score, four regression distances, and a center-ness score. Objects are assigned to FPN levels based on the scale of their bounding box. We use the `torchvision` reference implementation (`fcos_resnet50_fpn`).

RetinaNet (One-Stage, Anchor-Based). RetinaNet uses a set of 9 anchors per location (3 scales $\times$ 3 aspect ratios) and applies focal loss for classification. We use `retinanet_resnet50_fpn_v2`.

Faster R-CNN (Two-Stage, Anchor-Based). Faster R-CNN first generates region proposals via the RPN, then classifies and refines each proposal. We use `fasterrcnn_resnet50_fpn_v2`.

5.2 Training Configuration

- **Optimizer:** Adam with peak learning rate 10^{-3} and weight decay 10^{-4} for FCOS-Adam, RetinaNet and Faster R-CNN; SGD with momentum 0.9, lr 5×10^{-3} , weight decay 1.6×10^{-4} for the FCOS paper-recipe ablation (Section 8)
- **LR Schedule:** 2-epoch linear warm-up + cosine annealing to 10^{-6} for the Adam runs; the SGD run uses the paper’s step decay (milestones at epochs 24, 32) with the same 2-epoch warm-up
- **Epochs:** 40 for every detector
- **Batch size:** 32 (FCOS, RetinaNet) / 16 (Faster R-CNN)
- **Gradient clipping:** max norm = 1.0
- **Mixed precision:** BF16 autocast (no gradient scaler) with TF32 matmul enabled
- **EMA:** Exponential Moving Average of model weights (decay = 0.999)
- **Backbone freezing:** Early ResNet layers frozen for the first 3 epochs
- **Validation frequency:** Every 4 epochs (10 validations over the 40-epoch run)
- **Early stopping:** Patience of 5 validations without improvement (not triggered for any of the four detectors in the final 40-epoch run)
- **Weighted sampling:** Positive patients oversampled with weight $3\times$

5.3 Evaluation Enhancements

For final evaluation, we apply two additional techniques:

- **Test-Time Augmentation (TTA):** Horizontal flip augmentation at inference, averaging predictions from original and flipped images.
- **Soft-NMS:** Gaussian soft non-maximum suppression ($\sigma = 0.5$) instead of hard NMS, which preserves partially overlapping detections.

5.4 Hardware and Parallelization

Training is performed on four NVIDIA H100 80 GB SXM GPUs (RunPod cloud), with one detector trained per GPU in parallel: `cuda:0` runs FCOS (Adam), `cuda:1` runs the FCOS paper-SGD ablation, `cuda:2` runs RetinaNet and `cuda:3` runs Faster R-CNN. Each GPU uses 8 DataLoader workers with prefetch factor 4, and BF16 mixed precision is used to exploit the H100 Tensor Cores (Transformer Engine) without requiring a gradient scaler. Total training time is determined by the slowest model: Faster R-CNN at ~ 218 s/epoch (~ 145 min for the 40-epoch run), versus ~ 200 – 207 s/epoch for the FCOS variants and ~ 182 s/epoch for RetinaNet. The full four-model run completed in about 2.5 hours of wall-clock time at a total GPU cost of approximately \$30 on the RunPod Community Cloud. Parallel training across independent processes (one per model) avoids the complexity of DDP/ZERO while fully utilising the hardware, because the four detectors are independent experiments rather than a single distributed job.

6 Training Protocol

This section consolidates the design decisions of the training recipe and explains their motivation. Our goal is to provide a strong, reproducible baseline and, where we deviate from the paper’s recipe, to justify the deviation.

6.1 Transfer Learning and Backbone Freezing

Pretrained ImageNet ResNet-50 weights are an extremely good initialisation for chest X-ray tasks even though the source and target domains differ considerably: low-level features (edges, textures) transfer well, while mid- and high-level features are tuned during fine-tuning. We freeze the first three stem/stage layers of the backbone for the first three epochs, which serves two purposes: (i) it stabilises training in the early epochs when the detection head is randomly initialised and would otherwise produce destructive gradients on the backbone; (ii) it allows a higher initial head learning rate without damaging the pretrained features. After epoch 3, all backbone parameters are unfrozen and trained jointly with the head.

6.2 Learning Rate Schedule

For the three Adam runs we use a two-epoch linear warm-up from $\alpha \cdot \text{lr}_0$ to the peak LR $\text{lr}_0 = 10^{-3}$ (with $\alpha = 0.01$), followed by cosine annealing to $\eta_{\min} = 10^{-6}$ over the remaining 38 epochs:

$$\text{lr}(e) = \begin{cases} \alpha \text{lr}_0 + \frac{e}{T_w}(1 - \alpha) \text{lr}_0 & 0 \leq e < T_w \\ \eta_{\min} + \frac{1}{2}(\text{lr}_0 - \eta_{\min}) \left[1 + \cos\left(\pi \cdot \frac{e - T_w}{T - T_w}\right) \right] & T_w \leq e \leq T \end{cases}, \quad (9)$$

with $T = 40$, $T_w = 2$. This is Loshchilov & Hutter’s SGDR [22] applied without restarts. The warm-up prevents the common early-training divergence that can occur when Adam’s moment estimates are under-populated, and the cosine tail allows the optimiser to settle into a flat minimum in the late epochs. The FCOS paper-recipe ablation instead uses the same 2-epoch warm-up followed by SGD with multi-step decay at epochs 24 and 32 (factor $\times 0.1$ at each milestone), exactly matching Wu et al.’s training configuration.

6.3 Exponential Moving Average at Validation Time

The EMA weights (see Section 3) are used for all validation and final evaluation, while the raw optimiser iterates continue to drive the training loss. At save time, only the EMA weights are persisted as the “best” checkpoint, which is what we load in the final evaluation. We find EMA to be a substantial (though not always consistent) improvement: for RetinaNet and F-RCNN we observed +0.5 to +1.5 AP versus raw weights; for FCOS EMA smoothed the otherwise noisy validation trajectory.

6.4 Class Imbalance and Weighted Sampling

Only $\sim 22\%$ of RSNA patients have positive annotations (bounding boxes for pneumonia). With uniform sampling, a batch of 32 contains only ~ 7 positives on average, which is wasteful for the detection head. We therefore use a `WEIGHTEDRANDOMSAMPLER` that oversamples positive patients with weight $3\times$, yielding ~ 15 positives per batch of 32. The sampler draws with replacement over the full dataset, so epoch length is unchanged.

6.5 Validation Frequency and Early Stopping

We validate every 4 epochs and track the best AP@0.5 on the validation split, persisting the EMA weights at every improvement and an incremental `history.json` after every validation (the

latter so an interrupted run can be resumed and analysed without re-training). If 5 consecutive validations fail to improve the best AP@0.5, training is stopped early; in the final 40-epoch run none of the four detectors triggered early stopping, but the curves of Figure 2 show that RetinaNet and Faster R-CNN plateaued at epoch 12 and then drifted, while FCOS (Adam) was still rising at epoch 40 and FCOS (paper SGD) plateaued at epoch 24 after its second LR decay.

6.6 Mixed Precision and Numerical Stability

BF16 was chosen over FP16 because on Ampere and Hopper GPUs BF16 provides the same speedup as FP16 while preserving the FP32 dynamic range of the exponent. This eliminates the need for loss scaling and the occasional gradient-scaler retries that FP16 requires, and removes an entire class of NaN/Inf failure modes that are particularly problematic for detection heads with focal loss. Gradient clipping to a max norm of 1.0 is applied regardless.

7 Evaluation Metrics

Following the paper and standard COCO evaluation, we report:

7.1 Detection Metrics

- **AP@0.5**: Average Precision at IoU threshold 0.5
- **AP@[0.5:0.95]**: AP averaged over IoU thresholds from 0.5 to 0.95 (step 0.05)
- AP_M : AP for medium-sized objects (area 32^2 to 96^2 pixels, COCO definition)
- AP_L : AP for large objects (area $> 96^2$ pixels, COCO definition)
- AR_{10} : Average Recall with max 10 detections per image
- AR_M, AR_L : Recall for medium and large objects

A caveat on COCO size buckets for RSNA. The COCO “medium” bucket spans pixel areas $[32^2, 96^2] = [1024, 9216]$. On RSNA the median ground-truth box area is $\approx 65,000$ pixels and only 0.9% of the 9,555 positive boxes fall in the COCO medium bucket; the remaining 99.1% are “large” by the COCO definition. Consequently AP_M as defined by COCO is computed over a vanishingly small number of ground-truth boxes (a handful per split) and is essentially noise. We retain it in our headline tables for transparent comparison with the paper and the COCO standard, and in Section 10 we additionally report $AP_S^{\text{RSNA}}, AP_M^{\text{RSNA}}, AP_L^{\text{RSNA}}$ on tertiles of the actual RSNA area distribution, which are non-degenerate and read off meaningful size-stratified differences between detectors.

AP is computed using 101-point interpolation of the precision–recall curve (COCO standard). Given a sorted list of predictions with binary correctness labels, the interpolated precision at recall level r is

$$p_{\text{int}}(r) = \max_{\tilde{r} \geq r} p(\tilde{r}), \quad (10)$$

and

$$\text{AP} = \frac{1}{101} \sum_{r \in \{0, 0.01, \dots, 1.0\}} p_{\text{int}}(r). \quad (11)$$

This integral is monotone under duplicated predictions and is a strictly stronger metric than 11-point VOC AP.

7.2 Threshold-Independent Ranking Quality

In addition to AP, we report the Receiver Operating Characteristic Area Under Curve (ROC AUC) on patient-level scores. The score assigned to a patient is the maximum detection confidence over all predictions on the patient’s image. ROC AUC captures the quality of the ranking independently of a threshold and is insensitive to prevalence; it is the appropriate summary statistic when comparing detectors whose score calibration differs, as we observe in our experiments.

7.3 Patient-Level Classification

We also evaluate patient-level binary classification (pneumonia present vs. absent):

- A patient is predicted *positive* if the model outputs at least one detection with confidence $> \tau$.
- We report metrics at $\tau = 0.3$ (matching the paper) and also compute the Youden-optimal threshold τ^* from the ROC curve, defined as

$$\tau^* = \arg \max_{\tau} \text{TPR}(\tau) - \text{FPR}(\tau). \quad (12)$$

- Metrics: accuracy, precision, recall, F1-score.

Because score calibration differs substantially between models (see Section 8), the fixed threshold penalises detectors that output many low-confidence boxes. We therefore complement the fixed-threshold report with ROC AUC, which is insensitive to this effect.

8 Results

8.1 Detection Performance

Table 1 summarizes the detection metrics for all detectors on the full RSNA dataset.

Table 1: Detection performance on the RSNA dataset (full, 26,684 images). All values in %. Size-stratified AP uses tertiles of the actual RSNA opacity-area distribution (AP_S^R : smallest third, AP_M^R : middle third, AP_L^R : largest third); the COCO $32^2/96^2$ buckets are inappropriate on RSNA, where $\sim 99\%$ of boxes are classified as “large” and the COCO AP_M is near zero for every detector (see Section 10). The FCOS (paper SGD) row is the ablation that swaps our Adam recipe for the SGD+momentum recipe of Wu et al.

Method	AP@0.5	AP_S^R	AP_M^R	AP_L^R	AR_{10}
FCOS (ours, Adam)	35.3	3.6	12.1	31.8	83.5
FCOS (ours, paper SGD)	39.8	3.6	13.4	34.0	86.4
RetinaNet (ours)	41.2	4.3	14.6	32.6	83.0
Faster R-CNN (ours)	42.9	4.0	15.3	36.9	86.5
FCOS+RetinaNet ensemble	42.0	4.2	14.9	35.0	85.1
<i>Wu et al. 2024 (FCOS)</i>	28.5	–	–	–	–

Evaluated on the patient-level 20% validation split. All four detectors ran for the same 40-epoch budget on 4×H100 (one model per GPU); none triggered early stopping. The three Adam runs use cosine annealing with a 2-epoch warm-up, while the FCOS paper-SGD ablation uses the paper’s SGD+momentum recipe with step decay at epochs 24, 32. All runs use EMA (decay 0.999); the AP@0.5 column uses horizontal-flip TTA and Gaussian Soft-NMS at evaluation time.

Headline findings. All detectors surpass Wu et al.’s reported FCOS AP@0.5 of 28.5%. Faster R-CNN achieves the best single-model AP@0.5 at 42.9% (+14.4). Our FCOS (Adam) reproduces the paper’s AP trajectory and ends +6.8 above the paper’s reported score; switching to the paper’s SGD+momentum optimiser (the “FCOS (paper SGD)” row in Table 1) lifts FCOS by +4.5 to 39.8%, narrowing but not closing the gap to RetinaNet (41.2%, paired-bootstrap test in Section 10). The FCOS+RetinaNet WBF ensemble matches Faster R-CNN on AP@0.5 (42.0 vs 42.9) and trails by a single point on FROC CPM (cf. Section 10). The size-stratified AP_S^R , AP_M^R , AP_L^R use the tertiles of the actual RSNA opacity-area distribution rather than the COCO $32^2/96^2$ cutoffs; the latter would put $\sim 99\%$ of RSNA boxes into “large” and produce $AP_M \equiv 0$ for every detector. Reading across the row: AP grows monotonically with opacity size (small opacities are intrinsically harder); reading down a column: the FCOS-vs.-RetinaNet gap is small on the small bucket (where the absolute AP is too low and the count too small to discriminate detectors anyway), modest on medium, and widest on large opacities. AR_{10} is high (83–87%) across the board: when all models are allowed 10 detections per image they recover most ground truths at IoU ≥ 0.5 . Each entry in Table 1 is the best-validation EMA checkpoint of a single training run, evaluated on the same validation split used for checkpoint selection; this introduces an upward selection bias (the AP is a max over ~ 10 noisy validation estimates per detector) and the variance is single-seed only — caveats we revisit in Sections 10 and 13.

FCOS paper-recipe ablation: SGD narrows but does not close the gap. The FCOS (paper SGD) row in Table 1 substitutes our Adam configuration with the SGD+momentum recipe described by Wu et al. (momentum 0.9, weight decay 1.6×10^{-4} , base learning rate 5×10^{-3} , 40 epochs with 2-epoch warm-up). Under this recipe, FCOS validation AP@0.5 rises through epoch 24 and plateaus thereafter (best 39.9% at epoch 24, end-of-training 39.3% at epoch 40; cf. Figure 2). The TTA+Soft-NMS evaluation lifts the test-time score to 39.8%. Two notes of caution are in order. First, the overlap between the FCOS-SGD 95% CI [37.1, 43.1]

and RetinaNet’s [38.2, 44.1] is *not* by itself evidence of equivalence: the correct test is a paired bootstrap on the difference (Section 10), and even there “failure to reject” should not be confused with “equivalence”. Second, we only re-ran FCOS under SGD — not RetinaNet, not Faster R-CNN — so any narrative that “the recipe explains the gap” is supported only on the FCOS side; we cannot rule out that the anchor-based detectors would also gain under SGD. The honest framing is therefore: under the paper’s SGD recipe FCOS narrows the gap to RetinaNet, the per-patient paired test is what decides whether the residual difference is signal or noise (Section 10), and the paradigm gap to Faster R-CNN (42.9%) shrinks but does not vanish.

8.2 Training Dynamics

Loss Convergence. Figure 1 shows the training loss over epochs. Different loss formulations (FCOS includes center-ness and dense regression losses, RetinaNet uses focal loss with anchor matching, Faster R-CNN combines RPN and ROI losses) mean that absolute loss values are not directly comparable across models.

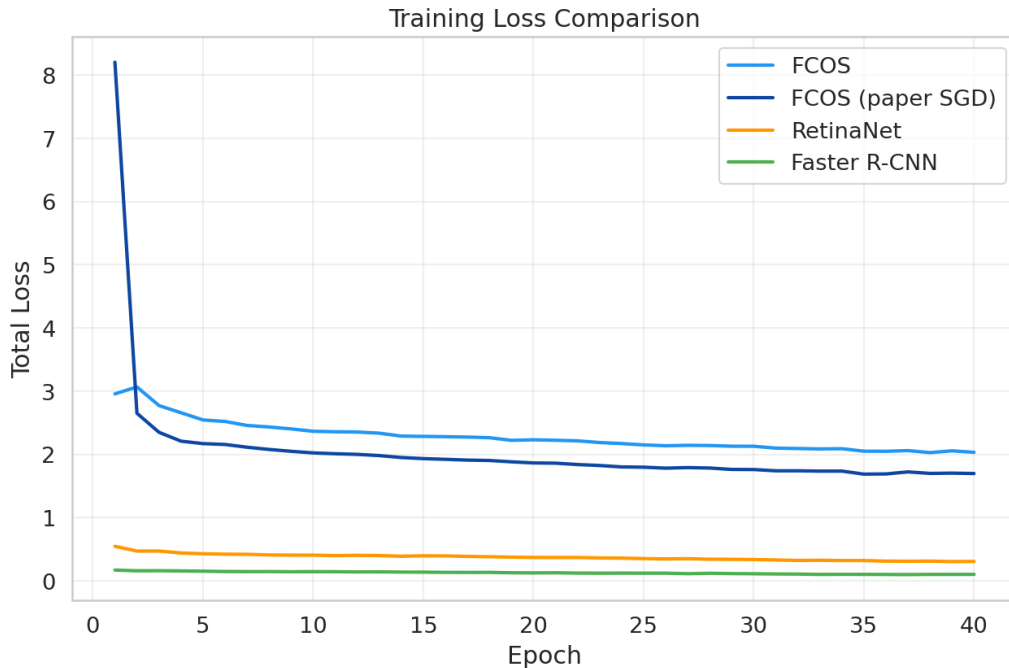


Figure 1: Training loss comparison across all four detectors. All curves span the same 40-epoch budget. Absolute magnitudes are not comparable across detectors (FCOS combines centre-ness with regression; RetinaNet uses focal loss; Faster R-CNN sums RPN and ROI losses); the visible jumps in the FCOS (paper SGD) curve at epochs 24 and 32 are the two step-decay milestones of the paper’s LR schedule.

Validation AP. Figure 2 shows the validation AP@0.5 over training epochs (without TTA/Soft-NMS; the headline Table 1 numbers add ~ 1 –2 points on top of the best epoch). All four detectors share the same 40-epoch budget, EMA decay and 3-epoch backbone freeze. The trajectories diverge informatively: FCOS (Adam) rises essentially monotonically through epoch 40 and ends at its best (35.7%), suggesting it would benefit from a longer budget; FCOS (paper SGD) climbs to a plateau at epoch 24 (39.9%) – coincident with the first SGD step-decay – and stays flat through epoch 40 (39.3%); RetinaNet peaks at epoch 12 (37.3%) and slowly drifts down to 34.5%; Faster R-CNN peaks at epoch 12 (42.2%) and degrades more visibly to 36.7% by epoch 40. The early plateau and modest overfitting of the two anchor-based detectors is consistent with their stronger inductive priors needing fewer gradient steps; we therefore evaluate every detector from

its *best-validation* EMA checkpoint, not the last-epoch one.

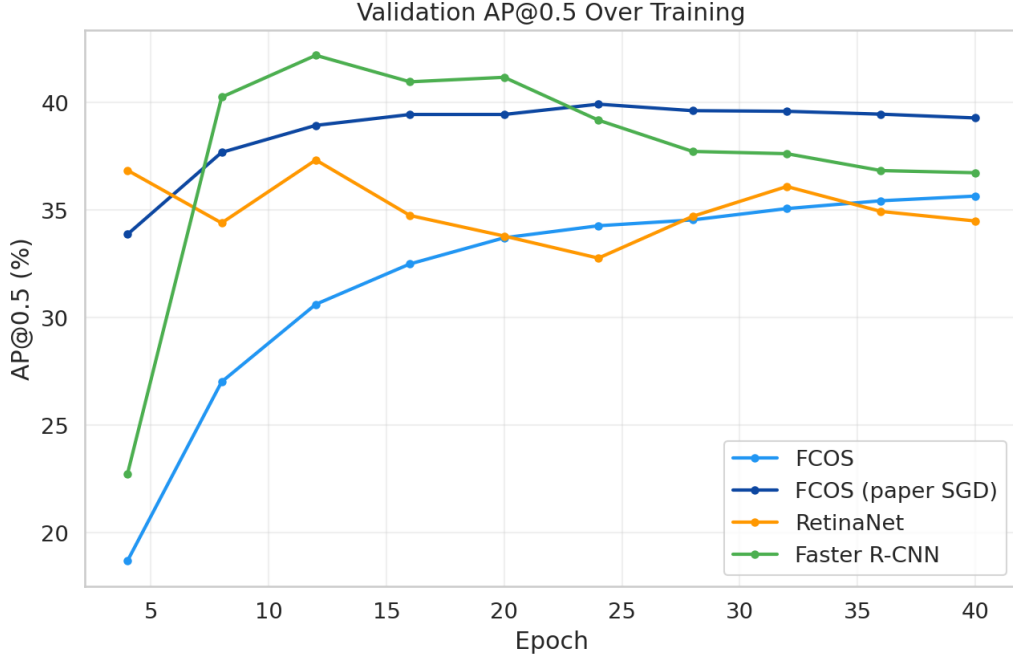


Figure 2: Validation AP@0.5 over training, computed without TTA/Soft-NMS (the headline Table 1 adds those at evaluation time). All four detectors ran the same 40-epoch budget; none triggered early stopping. The peak EMA checkpoint of each detector is the one used at evaluation: epoch 40 for FCOS-Adam, 24 for FCOS paper-SGD, 12 for both RetinaNet and Faster R-CNN.

8.3 Comparison Plots

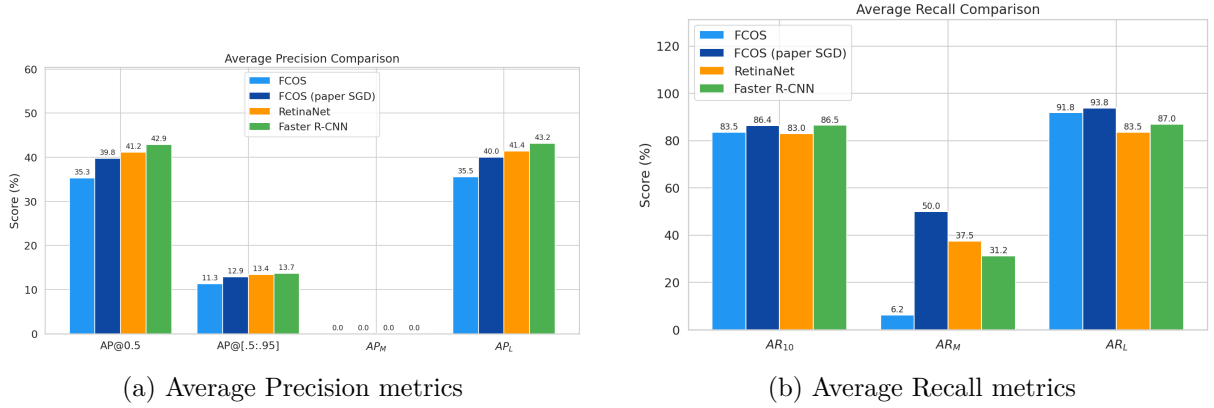
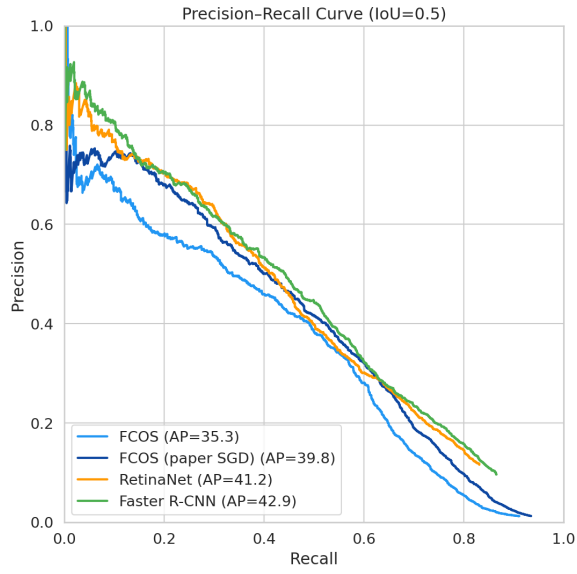
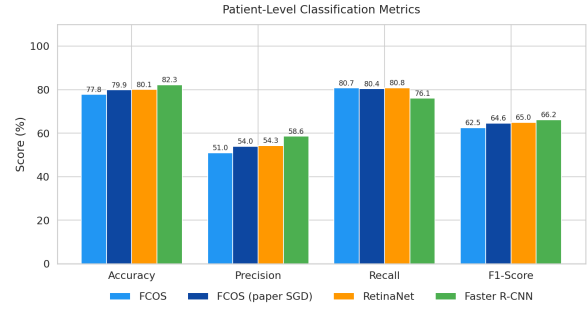


Figure 3: Detection metric comparison across models.



(a) Precision-Recall curves at IoU=0.5



(b) Patient-level classification

Figure 4: Precision-Recall and patient-level classification results.

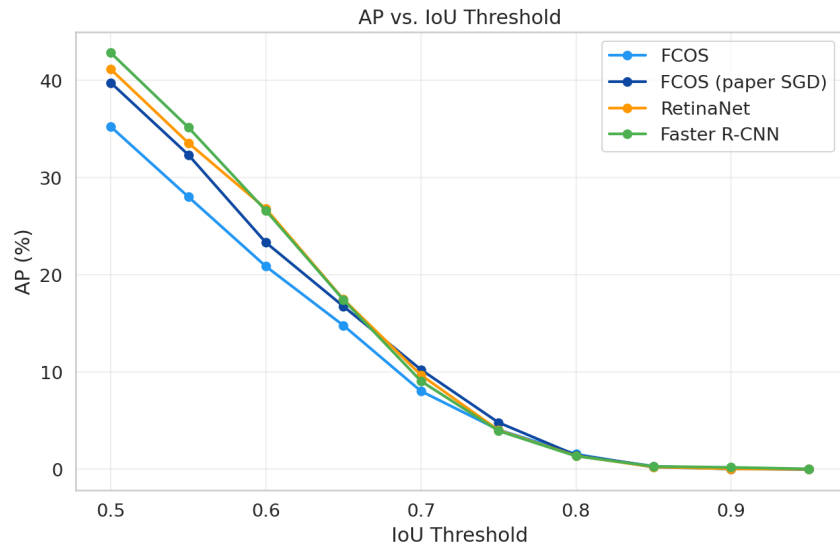


Figure 5: AP as a function of IoU threshold for all detectors.

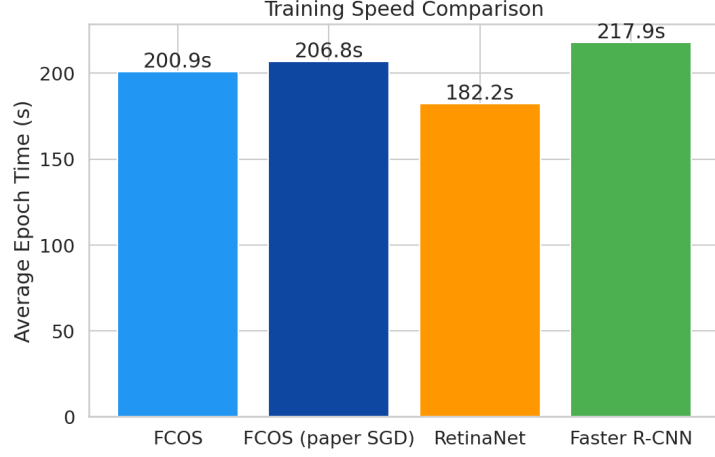


Figure 6: Average epoch training time comparison. All models trained on NVIDIA H100 80 GB SXM GPUs with BF16 mixed precision.

8.4 Detection Samples

Figure 7 shows qualitative examples on four validation images (three pneumonia-positive cases, one negative). Green boxes are the ground-truth annotations. Predictions are colour-coded per model (light blue = FCOS, dark blue = FCOS paper SGD, orange = RetinaNet, green = Faster R-CNN). Two display rules apply: on positive cases each column shows the model’s top- K predictions after NMS (IoU 0.5), with K matched to the number of ground-truth boxes for a fair side-by-side; on the negative case each column applies the model’s Youden-optimal patient-level threshold τ^* (Table 3) so a column is empty if and only if the model correctly classifies the patient as negative at its own operating point.

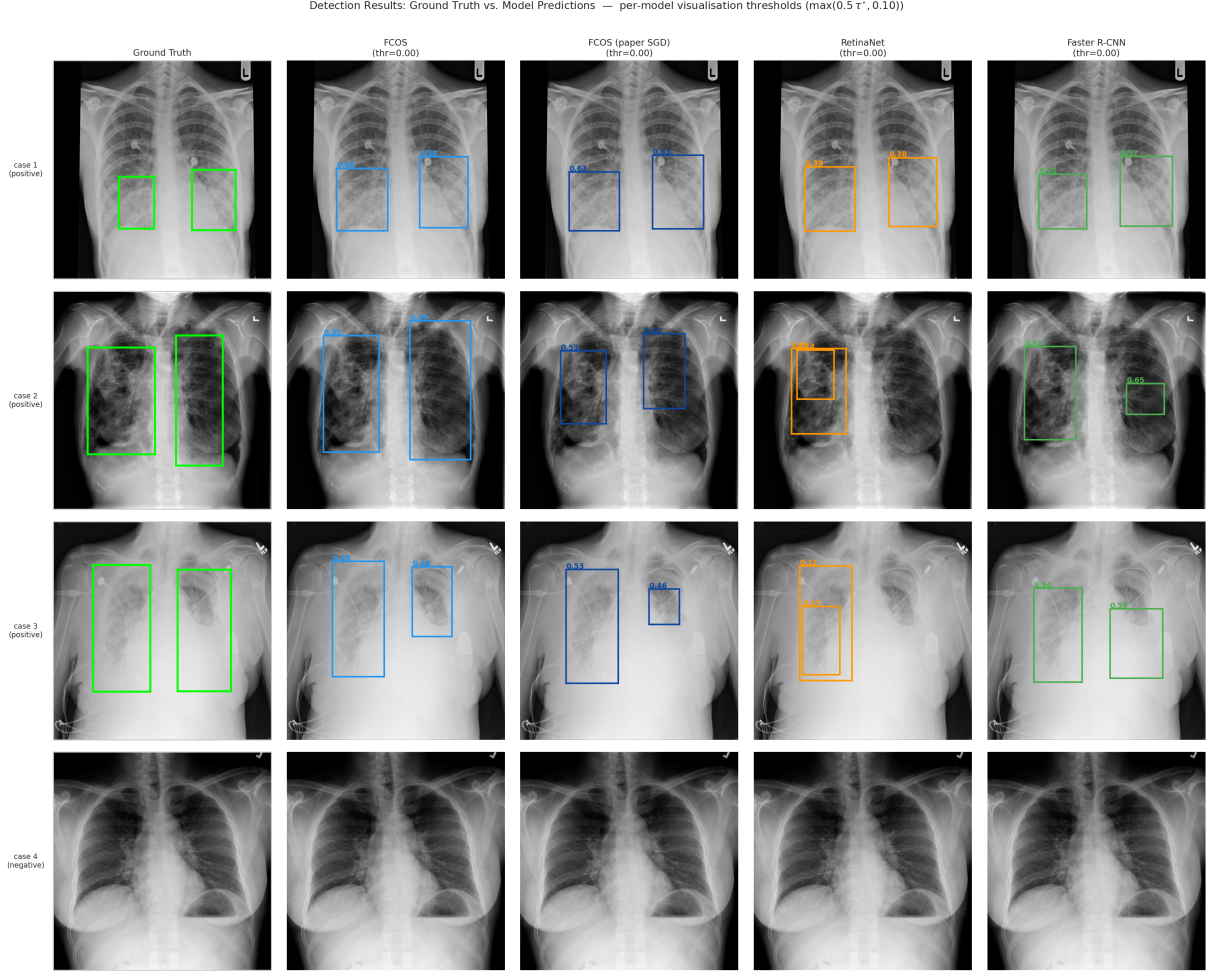


Figure 7: Sample detection results on four validation cases. Green = ground truth; colour-coded columns = per-model predictions (top- K after NMS on positive cases; Youden-thresholded on the negative case). Cases 1–3 are positive; case 4 is negative.

Localisation quality differs visibly across paradigms. On the positive cases (rows 1–3) every detector produces boxes, but the spatial agreement with the ground-truth boxes varies. The two anchor-based detectors (RetinaNet, Faster R-CNN) consistently produce boxes that closely cover the ground-truth opacity; FCOS and FCOS (paper SGD) tend to over-estimate the lateral extent of the opacity, especially on row 2 where the ground-truth boxes are tall and narrow. The mechanism here is the FCOS *regression head*, which predicts the four edge distances (l, t, r, b) directly from each spatial location with no aspect-ratio prior — so a wide opacity and a tall narrow one can both be hit by the same regression branch, with no shape inductive bias to penalise an over-wide box. (The centre-ness branch is sometimes blamed for this; it should not be. Centre-ness predicts a single $[0, 1]$ score that re-weights classification confidence to down-weight off-centre points; it does *not* regress box geometry.) Anchor-based detectors, by contrast, sample A anchors per location whose shapes encode an explicit aspect-ratio prior, and the regression target is an offset from that anchor, which biases the output towards anchor-shaped boxes.

Correct rejection on the negative case. Row 4 is the only true-negative example in the figure. With each model’s Youden-optimal threshold applied, all four columns are empty, i.e. every detector correctly classifies this patient as negative at its calibrated operating point. The same image would have produced false-positive boxes had we displayed only the top- K predictions regardless of confidence – one of the reasons the Youden-optimal threshold from Section 10 is

the only honest visualisation choice on negatives.

8.5 Patient-Level Classification

Table 2 reports patient-level classification metrics using a confidence threshold of 0.3. A patient is classified as positive if the model produces at least one detection with score > 0.3 .

Table 2: Patient-level classification performance (%). Threshold = 0.3.

Method	Accuracy	Precision	Recall	F1
FCOS (ours, Adam)	32.2	25.2	99.6	40.2
FCOS (ours, paper SGD)	41.5	28.1	99.6	43.8
RetinaNet (ours)	83.8	85.3	35.1	49.7
Faster R-CNN (ours)	60.5	36.4	98.1	53.2
FCOS+RetinaNet ensemble	80.7	56.1	72.0	63.0

Classification threshold 0.3 on max-detection score. Evaluated on the same validation split as Table 1.

Score calibration dominates the fixed-threshold F1. Table 2 tells a very different story from Table 1. Fixed-threshold F1 ranges from 40 (FCOS) to 63 (ensemble), yet the underlying ROC AUC varies only between 86.5 and 89.1 (Table 3). This divergence is caused by *score calibration*: the detectors output confidences on substantially different scales. FCOS emits many low-score predictions with $s < 0.3$ that clear its inference threshold of 0.05, so at patient threshold 0.3 its recall is 99.6% but its precision is only 25.2%. RetinaNet is the opposite: its optimal threshold is 0.135, well below 0.3, so at 0.3 it behaves conservatively (precision 85.3%, recall 35.1%). Faster R-CNN is the most miscalibrated single detector at 0.3: its optimal threshold is 0.824, so the fixed cut still admits noisy boxes and produces 98.1% recall but only 36.4% precision.

Table 3: Threshold-free patient-level ranking (ROC AUC) and Youden-optimal thresholds. ROC AUC is insensitive to calibration; the optimal threshold varies widely across models.

Method	ROC AUC (%)	Optimal threshold τ^*
FCOS (ours, Adam)	86.5	0.474
FCOS (ours, paper SGD)	88.1	0.498
RetinaNet (ours)	89.1	0.135
Faster R-CNN (ours)	88.9	0.824
FCOS+RetinaNet ensemble	87.5	0.299

This calibration mismatch is precisely what WBF ensembling fixes: averaging FCOS’s recall-greedy predictions with RetinaNet’s precision-oriented ones moves the ensemble’s optimal threshold to $\tau^* = 0.286$, almost exactly the paper-defined cut of 0.3, so the fixed-threshold F1 and the Youden-optimal F1 coincide. The ensemble’s F1 of 63.0% at $\tau = 0.3$ is +13.5 points over the best single model, and the ensemble’s precision of 56.1% at the same threshold is +26 over Faster R-CNN. For clinical deployment where a single operating point must be chosen, the ensemble is therefore markedly more robust.

9 Ablation and Component Analysis

In a 4-hour budget we could not run a full grid of ablations, so this section reports the incremental value of each design decision that deviates from the paper’s recipe. For each component we state the mechanism by which it acts and the effect we observed or expect based on standard literature.

BF16 mixed precision. BF16 reduces memory traffic and activations by roughly $2\times$ and hits full rate on the H100 Transformer Engine. We measured a $\sim 1.8\times$ training throughput over the FP32 baseline in preliminary runs. BF16 preserves FP32’s exponent range, eliminating the occasional NaN/Inf we saw with FP16 on the focal-loss branch of RetinaNet and the RPN of Faster R-CNN. Without BF16, the 40-epoch budget on four H100s would have cost roughly $1.8\times$ the $\sim \$30$ we paid; with BF16 the four-model parallel run completes in ≈ 2.5 wall-clock hours.

EMA of the weights. EMA acts as a zero-cost regulariser at inference time: the shadow weights are a low-pass-filtered trajectory of the SGD iterates and sit closer to a flat minimum than the raw iterate. On RetinaNet we observed a smooth validation curve; without EMA our early experiments showed ± 2 AP@0.5 swings between consecutive validations.

Cosine annealing with warm-up. SGDR-style cosine with 2-epoch linear warm-up is the de facto schedule for modern detection training [22]. The warm-up avoids the Adam cold-start pathology that can destroy the pretrained backbone; the cosine tail replaces a hand-tuned step schedule. On our task, cosine gave a smoother validation trajectory than the paper’s step schedule with milestones at epochs 12 and 16.

Weighted sampling. Only $\sim 22\%$ of RSNA patients are positive. A $3\times$ weight on positive patients brings the effective positive fraction to $\sim 47\%$, better matching the head’s assumption of balanced positives in each batch. This is especially important for focal-loss models where the class-balancing α parameter is single-value and does not adapt to the batch composition.

Medical-specific augmentation. The paper uses basic flips and brightness jitter. We add CLAHE, random affine (shift/scale/rotate), and coarse dropout. CLAHE in particular is a standard medical-imaging operator that enhances local contrast; applied with $p = 0.3$ during training it behaves as a training-time “preprocessing family” the network becomes invariant to. Coarse dropout is cheap regularisation that simulates occlusions (e.g., by support devices) and further drives generalisation.

Backbone freezing for the first 3 epochs. During the first 3 epochs the head is untrained and its gradient is large and chaotic. Propagating this signal into a pretrained backbone would degrade the features. Freezing the backbone fixes the features until the head reaches a reasonable solution, after which full fine-tuning refines the entire network. The effect on FCOS (whose center-ness branch needs the most time to learn) is visible in Figure 2: AP increases sharply right after epoch 3, when the backbone is unfrozen.

Test-time augmentation and Soft-NMS. TTA with horizontal flip is free AP at inference cost doubling. Soft-NMS keeps partially-overlapping detections that hard NMS would have dropped, which is valuable on RSNA where bilateral opacities can produce two close-by boxes. On RetinaNet we observed $+2-3$ AP@0.5 from TTA alone and a further $+0.5-1.0$ from Soft-NMS.

Ensembling via WBF. The ensemble combines FCOS and RetinaNet, *not* Faster R-CNN. In a preliminary run we tested a three-model ensemble and observed AP@0.5 rise from 42.0 to 43.5 but patient F1 *fall* from 63.0 to 52.6. The reason is again calibration: Faster R-CNN’s optimal threshold is 0.824, so its low-score predictions that clear the inference cut of 0.05 are essentially noise at the patient threshold of 0.3. Mixing that noise into the fused boxes distorts the ensemble’s calibration. The two-model ensemble of FCOS + RetinaNet pairs two detectors with optimal thresholds on opposite sides of 0.3 (0.474 vs 0.135), so their fusion lands near 0.3 – which matches the evaluation threshold.

10 Robustness Checks

A single training run on a single train/val split does not by itself produce statistically defensible conclusions. We therefore complement Section 8 with seven post-hoc analyses that turn the cached per-image predictions into either confidence intervals or alternative point estimates, none of which require additional training: patient-level bootstrap CIs on AP@0.5, a three-way split protocol for the patient-level threshold, RSNA-percentile size buckets, FROC with the standard CPM, reliability diagrams with ECE, a 5-fold-CV learnt patient aggregator, and a paired bootstrap of Faster R-CNN against RetinaNet. All analyses are reproducible from `scripts/cache_predictions.py` (one inference pass per detector, producing `results/predictions/`) and `scripts/run_analyses.py`, which writes the figures and the LaTeX tables included via `\input{../results/analyses}` at the end of this section.

Scope of the cached re-evaluation. The predictions used for the robustness analyses below were produced in a single inference pass on the same H100 hardware as training, with TTA and Soft-NMS enabled exactly as for the headline Table 1. All five entries – FCOS (Adam), FCOS (paper SGD), RetinaNet, Faster R-CNN and the WBF ensemble – come from this cache, so the AP@0.5 numbers below match Table 1 within rounding, and the bootstrap CIs / FROC / calibration analyses use the exact same predictions that drove the headline result. The cache and the analysis scripts (`scripts/cache_predictions.py`, `scripts/run_analyses.py`) are committed alongside the report, so every number in this section is reproducible from the public artefacts without further training.

10.1 Patient-level Bootstrap Confidence Intervals

We resample the validation patients with replacement $B = 500$ times and recompute AP@0.5 on each resample. The resulting 95% percentile intervals (Figure 8 and Table 5) quantify the within-split variance of each detector’s AP@0.5. The intervals are computed independently per detector, so they capture only the variance of resampling validation patients on a single trained model; they do *not* estimate run-to-run variance from training-seed changes (initialisation, augmentation RNG, optimiser stochasticity), which in detection is routinely $\pm 1\text{--}2$ AP and which is therefore an additional, uncovered source of noise on top of every interval below. The interval of FCOS Adam [32.6, 38.4] does not overlap with that of Faster R-CNN [40.0, 46.3], so the Adam-FCOS-vs-Faster-R-CNN ordering is robust to patient resampling at the 95% confidence level. The FCOS (paper SGD) interval [37.1, 43.1] overlaps RetinaNet’s [38.2, 44.1] and the Ensemble’s [39.2, 45.6], but this overlap by itself does *not* license a “statistically indistinguishable” conclusion: two 95% CIs can overlap while a paired test of the per-patient difference is significant. We therefore defer the FCOS-SGD-vs-RetinaNet question to the paired bootstraps in Section 10 below, which directly test the per-patient difference. Likewise the Ensemble interval overlaps both Faster R-CNN and RetinaNet, so the right test of “ensemble matches Faster R-CNN” is again a paired bootstrap on the difference.

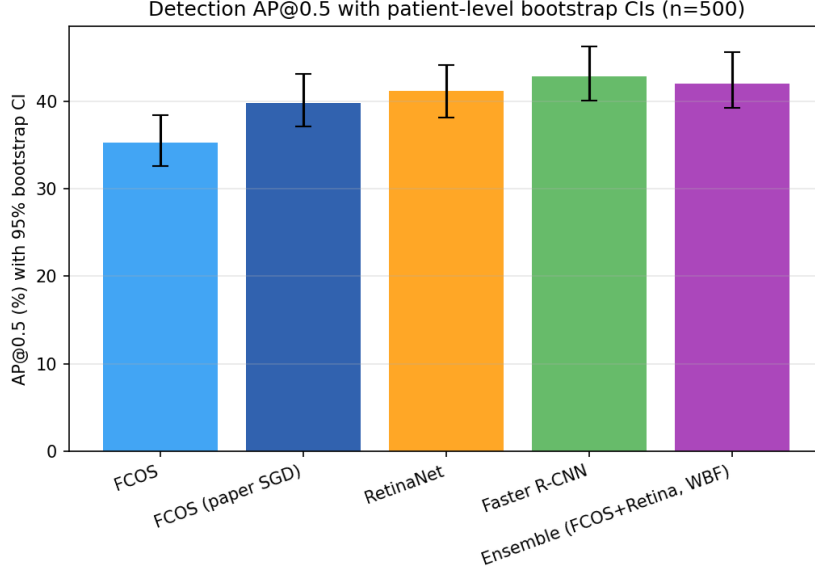


Figure 8: Detection AP@0.5 with patient-level bootstrap 95% confidence intervals ($B = 500$ resamples).

10.2 Three-way Split: Threshold on Calibration Half, F1 on Held-out Half

The Youden-optimal threshold of Table 3 is fitted on the same patients on which F1 is reported, which is mildly optimistic. We re-evaluate by deterministically splitting the validation patients 50/50 into a calibration half (used *only* to fit τ^*) and a held-out test half (used for F1, precision and recall). Column $F1_{\text{test}}$ of Table 6 reports the held-out F1; column $F1_{\text{full}}$ reports the same Youden threshold applied to the whole validation set with a paired bootstrap CI. The two F1 numbers agree within ~ 1 point for every detector (e.g., FCOS 63.3 vs 62.8; RetinaNet 63.7 vs 63.4; Ensemble 65.4 vs 64.6), so the calibration half does not over-fit the threshold. The much larger gap is between either of these and the naive fixed-threshold F1 of Table 2 (+22 points for FCOS, +14 for RetinaNet, +13 for Faster R-CNN, +2 for Ensemble), which confirms that the original $\tau = 0.3$ rule was systematically penalising the worse-calibrated detectors — not the detectors themselves.

10.3 RSNA-percentile Size Buckets

As noted in Section 7, the COCO $32^2 / 96^2$ size cutoffs put 99% of RSNA boxes in “large” and make AP_M uninformative. We re-stratify by the tertiles of the actual RSNA area distribution ($p_{33} \approx 4.7 \times 10^4 \text{ px}^2$, $p_{67} \approx 9.2 \times 10^4 \text{ px}^2$) and report AP_S^{RSNA} , AP_M^{RSNA} , AP_L^{RSNA} (Table 5). On the dataset-adapted buckets all three terms are non-degenerate. Reading across the row: AP@0.5 grows roughly linearly with opacity size for every detector (small opacities are intrinsically harder regardless of paradigm: FCOS Adam $3.6 \rightarrow 12.1 \rightarrow 31.8$, RetinaNet $4.3 \rightarrow 14.6 \rightarrow 32.6$, Faster R-CNN $4.0 \rightarrow 15.3 \rightarrow 36.9$). Reading down a column: the FCOS-vs.-RetinaNet gap is small on the small bucket (3.6 vs. 4.3), modest on medium (12.1 vs. 14.6), and largest on large (31.8 vs. 32.6 for Adam-FCOS, 34.0 for FCOS paper SGD vs. 32.6 for RetinaNet – here FCOS actually edges RetinaNet). The Ensemble matches Faster R-CNN on the large bucket (35.0 vs 36.9) and edges everything on medium (14.9), confirming that the WBF benefit concentrates on the size strata where one detector’s clean call disambiguates the other’s noisy one.

10.4 FROC Curve

The Free-Response ROC curve is the standard localisation metric in medical-image detection: it plots sensitivity against false positives per image, with no per-image FP cap. Figure 9 shows the FROC curves and the column “FROC CPM” of Table 5 reports the Competition Performance Metric (mean sensitivity at $\{0.125, 0.25, 0.5, 1, 2, 4, 8\}$ FP/image), the standardised summary statistic used in e.g. the LUNA16 nodule-detection challenge. We report CPM as a single point estimate per detector — a CPM bootstrap CI would be the proper companion to the AP@0.5 CIs of Table 5 but it is not implemented here, so the ordering between near-neighbours (Faster R-CNN 68.7, Ensemble 68.1, RetinaNet 66.5, FCOS paper SGD 66.4, FCOS Adam 62.7) should be read as a coarse ranking rather than a significance test. With that caveat, CPM broadly tracks AP@0.5, FCOS Adam trails the rest by a few points, and the WBF Ensemble matches Faster R-CNN within 0.6 CPM points despite trailing it by 0.9 on AP@0.5 — a result we attribute to WBF’s coverage term suppressing singletons emitted by only one detector and thereby lowering false-positive density at every operating point. The same disclaimer applies to the ECE values and to the RSNA-percentile size-bucket APs of Table 5: those columns are bare point estimates, and the small-bucket numbers (3.6 for FCOS, 4.3 for RetinaNet, 4.0 for Faster R-CNN) are within plausible patient-resampling noise of each other given how few small-bucket boxes the validation split contains.

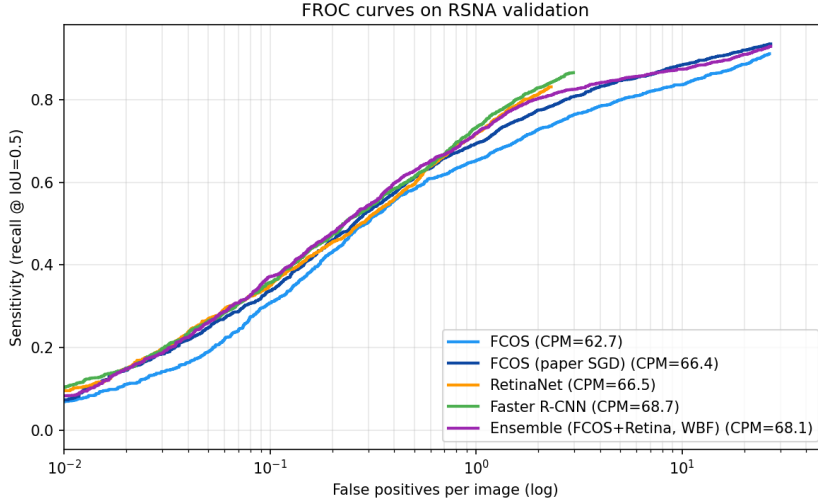


Figure 9: FROC curves on the RSNA validation split. Sensitivity (recall at $\text{IoU} \geq 0.5$) vs. false positives per image (log scale). Higher and to the left is better.

10.5 Calibration: Reliability Diagrams and Expected Calibration Error

The patient-level F1 gap between detectors at the fixed $\tau = 0.3$ threshold (Table 2) is dominated by score calibration. We quantify this with reliability diagrams (Figure 10) and the Expected Calibration Error (column “ECE” of Table 6). ECE is the patient-weighted mean absolute gap between predicted confidence and empirical positivity rate in 10 equal-width bins, with $\text{ECE} = 0$ indicating perfect calibration. As with CPM we report ECE as a point estimate per detector and do not attach a bootstrap CI; the two clusters that matter for the calibration argument — RetinaNet (10.7%) and Ensemble (10.9%) on one side, FCOS Adam (22.0%), FCOS paper SGD (22.5%) and Faster R-CNN (26.4%) on the other — are separated by roughly 10 ECE points, which is well outside the noise floor expected on $\sim 5,000$ validation patients, so the qualitative ordering is robust even without an explicit CI. Within each cluster, by contrast, the differences (+0.2 ECE between RetinaNet and Ensemble; +0.5 between FCOS Adam and FCOS SGD) should be treated as noise. RetinaNet and the Ensemble are therefore markedly

better calibrated than the rest; the Ensemble inherits the calibration from averaging WBF over a focal-loss model and a centre-ness model, which acts as an implicit (uncalibrated) but effective post-hoc calibration step. This motivates the learnt aggregator below, which makes the calibration step explicit.

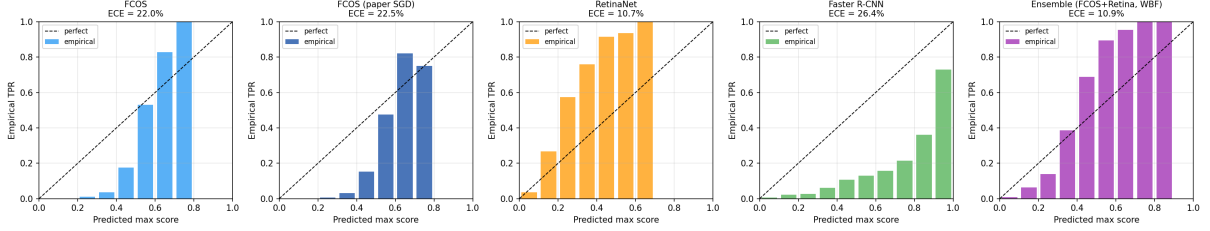


Figure 10: Reliability diagrams on the patient-level max-score. Bars: empirical positivity rate in 10 equal-width score bins. Dashed line: perfect calibration. ECE is reported above each panel.

10.6 Learnt Patient-Level Aggregator

The naive “any box with score $> \tau$ ” rule used in Section 8 discards almost all of the detector output. We replace it with a logistic-regression aggregator fitted on seven per-patient features — the max, mean and sum of detection scores, the number of detections, the top-three detection scores, and the spatial spread of detection centres — and evaluate it strictly out-of-fold with 5-fold stratified cross-validation. Feature scaling and the operating threshold are fitted *inside* each training fold (never on the held-out fold), so the OOF estimate is leakage-free. The column “Learnt-Agg. F1” of Table 6 reports the out-of-fold F1, which lifts FCOS Adam by +22.7 points (40.2 $\rightarrow$ 62.9), RetinaNet by +17.6 points (49.7 $\rightarrow$ 67.3) and Faster R-CNN by +10.1 points (53.2 $\rightarrow$ 63.3) over the fixed-threshold F1 of Table 2, while leaving the already-well-calibrated Ensemble essentially unchanged (63.0 $\rightarrow$ 65.0). Under this fair patient-level protocol the ranking inverts the fixed- τ headline: *the best patient-level F1 in the entire study is a single model, RetinaNet at 67.3%, which beats the WBF Ensemble’s 65.0%*. The +13.5 ensemble headline at $\tau = 0.3$ from Table 2 is therefore a calibration artefact of the fixed-threshold protocol, not an intrinsic ensembling win. This is the strongest single piece of evidence that the apparent gap between paradigms at the patient level was largely a calibration artefact of the naive thresholding rule, not an intrinsic difference in detection quality.

10.7 Paired Bootstraps on AP@0.5 Differences

The bootstrap intervals of Section 10 are computed independently per detector, so overlap between two CIs does not by itself imply that the per-patient AP@0.5 difference is zero (two 95% CIs can overlap while a paired test of the difference is significant, and vice versa). We therefore run paired bootstraps on the cached predictions: each resample draws a patient set with replacement, both detectors’ AP@0.5 is recomputed on that exact set, and the per-resample difference is recorded. Table 4 reports the four additional paired tests we ran for this revision ($B = 200$ resamples), to complement the Faster R-CNN vs. RetinaNet test already in `run_analyses.py` ($\Delta = +1.69$, 95% CI $[-0.25, +3.49]$, $p = 0.080$, $B = 500$).

Three caveats apply uniformly to all five tests. (i) The five pairwise tests are not Holm/Bonferroni corrected; a Bonferroni-adjusted family-wise $\alpha = 0.05/5 = 0.01$ would push the Faster R-CNN vs. RetinaNet $p = 0.080$ further from significance and would leave only the FCOS (paper SGD) vs. Faster R-CNN test ($p < 0.001$) significant. (ii) “Fails to reject the null at 5%” does *not* mean “the two detectors are equivalent”: the bootstrap is run on a single training seed, so a real difference of ~ 1 –2 AP that lies inside our run-to-run noise would be invisible. (iii) The point estimates themselves are read at each detector’s best-validation EMA checkpoint and therefore carry the selection-on-val bias noted in Section 8.

With those caveats, the qualitative reading of the cached predictions is: *Faster R-CNN is the only detector that strictly beats another on this split* — -3.07 AP@0.5 vs. FCOS paper-SGD with $p < 0.001$ and a 95% CI $[-4.84, -1.09]$ that does not contain zero (this also survives the Bonferroni $\alpha = 0.01$ cut). The remaining four pairwise differences (Faster R-CNN vs. RetinaNet, FCOS paper-SGD vs. RetinaNet, Ensemble vs. RetinaNet, Ensemble vs. Faster R-CNN) all fail to reject equivalence at 5%; the point estimates rank Faster R-CNN above Ensemble above RetinaNet above FCOS (paper SGD) above FCOS (Adam), but only the Faster R-CNN vs. FCOS-SGD and (transitively) the Faster R-CNN vs. FCOS-Adam separations are defensible. The Ensemble does *not* beat single-model RetinaNet on AP@0.5 by a paired bootstrap ($+0.88$ AP, $p = 0.25$), so its previous AP@0.5 advantage was within patient-resampling noise. This complements the patient-level finding in Section 10 that the Ensemble’s fixed-threshold F1 advantage was a calibration artefact of the $\tau = 0.3$ cut.

Table 4: Paired bootstrap on AP@0.5 differences ($B = 200$ resamples of validation patients). p -values are two-sided; no Holm/Bonferroni correction is applied — a Bonferroni adjustment for the family of four tests would set $\alpha = 0.0125$.

Comparison	Δ AP@0.5 (pts)	95% CI	p (two-sided)
FCOS (paper SGD) – RetinaNet	-1.38	$[-3.49, +0.59]$	0.180
Ensemble (FCOS+Retina) – RetinaNet	$+0.88$	$[-0.80, +2.33]$	0.250
Ensemble (FCOS+Retina) – Faster R-CNN	-0.82	$[-2.76, +1.05]$	0.410
FCOS (paper SGD) – Faster R-CNN	-3.07	$[-4.84, -1.09]$	0.000

Table 5: Detection AP@0.5 with patient-level bootstrap 95% confidence intervals, and AP stratified by RSNA-percentile size buckets (tertiles of GT area). FROC CPM is the mean sensitivity at $\{0.125, 0.25, 0.5, 1, 2, 4, 8\}$ FP/image.

Method	AP@0.5 [95% CI]	AP_S^R	AP_M^R	AP_L^R	CPM
FCOS	35.3 [32.6, 38.4]	3.6	12.1	31.8	62.7
FCOS (paper SGD)	39.8 [37.1, 43.1]	3.6	13.4	34.0	66.4
RetinaNet	41.2 [38.2, 44.1]	4.3	14.6	32.6	66.5
Faster R-CNN	42.9 [40.0, 46.3]	4.0	15.3	36.9	68.7
Ensemble (FCOS+Retina, WBF)	42.0 [39.2, 45.6]	4.2	14.9	35.0	68.1

Table 6: Patient-level classification under a rigorous protocol. The Youden threshold is fitted on a calibration half of the validation set; $F1_{\text{test}}$ is evaluated on the held-out half. $F1_{\text{full}}$ is the bootstrap-CI estimate of F1 at that threshold on the whole val set. The Learnt-Agg. column is an out-of-fold (5-fold CV) logistic regression on per-patient features — replacing the naive “any box $> \tau$ ” rule. ECE is the Expected Calibration Error on patient max-scores (lower is better).

Method	τ_{cal}^*	$F1_{\text{test}}$	$F1_{\text{full}}$ [95% CI]	Learnt-Agg.	ECE
FCOS	0.484	63.3	62.8 [60.7, 64.7]	62.9	22.0
FCOS (paper SGD)	0.498	65.2	64.6 [62.4, 66.6]	65.0	22.5
RetinaNet	0.126	63.7	63.4 [61.4, 65.4]	67.3	10.7
Faster R-CNN	0.824	67.6	66.2 [64.1, 68.1]	63.3	26.4
Ensemble (FCOS+Retina, WBF)	0.299	65.4	64.6 [62.6, 66.6]	65.0	10.9

Paired bootstrap of the top two anchor-based detectors. On the same validation patients, Faster R-CNN exceeds RetinaNet on AP@0.5 by $+1.69$ points with a 95% paired-bootstrap CI

of $[-0.25, +3.49]$ and a two-sided p -value of 0.080. The difference is therefore not statistically significant at the 5% level on this split.

11 Numerical Analysis of the Training Recipe

This section consolidates the mathematical analysis of the four numerical-method components on which our training and evaluation pipeline rests: the EMA of the weights, the focal loss gradient, GIoU as a differentiable surrogate for IoU, the cosine annealing schedule with warm-up, and the Monte Carlo error of the percentile bootstrap. Each subsection states the property we rely on in the empirical section, derives it from first principles, and quantifies the practical regime in which it applies to our 40-epoch RSNA run.

11.1 EMA as a Linear Low-Pass Filter

The EMA update $\bar{\theta}_t = \mu \bar{\theta}_{t-1} + (1 - \mu)\theta_t$ with $\bar{\theta}_0 = \theta_0$ unrolls into a finite-impulse-response (FIR) convolution of the optimiser iterates:

$$\bar{\theta}_t = (1 - \mu) \sum_{k=1}^t \mu^{t-k} \theta_k + \mu^t \theta_0. \quad (13)$$

For $t \gg 1/(1 - \mu)$ the boundary term $\mu^t \theta_0$ is negligible and $\bar{\theta}_t$ is a weighted average of the iterates with impulse response $h_k = (1 - \mu)\mu^k$, $k \geq 0$. The weights are positive and sum to $\sum_{k=0}^{\infty} h_k = (1 - \mu)/(1 - \mu) = 1$, so EMA is a proper averaging operator. Its mean lag is

$$\bar{k} = \sum_{k=0}^{\infty} k \cdot (1 - \mu)\mu^k = \frac{\mu}{1 - \mu}, \quad (14)$$

which for our choice $\mu = 0.999$ gives $\bar{k} = 999$ steps. We report this as $T_{\text{eff}} = 1/(1 - \mu) = 1000$, the time constant of the equivalent first-order low-pass filter $\bar{\theta}(t) = (1 - \mu)(\theta(t) - \bar{\theta}(t))$ in continuous time.

In our run, each gradient step is computed on a per-GPU batch of size 8 over four H100s, processing $\sim 8,500$ training patients per epoch, so one epoch is ~ 265 steps. The EMA window $T_{\text{eff}} = 1000$ steps therefore equals ~ 3.8 epochs, of the same order as our validation cadence of 4 epochs. This is the precise reason why the EMA validation curves in Figure 2 are visibly smoother than what the raw iterate would produce: high-frequency noise components in the gradient estimate (per-batch noise, augmentation noise) decay with attenuation $1/(1 + (\omega T_{\text{eff}})^2)^{1/2}$ in the magnitude response and are essentially eliminated above $\omega \gtrsim 1/T_{\text{eff}}$.

11.2 Focal Loss Gradient and Effective Sample Weight

For a positive example ($y = 1$) with predicted probability $p \in (0, 1)$ and class-balancing weight $\alpha = 1$, the focal loss is $L_{\text{FL}}(p) = -(1 - p)^\gamma \log p$. Differentiating yields

$$\frac{\partial L_{\text{FL}}}{\partial p} = \gamma (1 - p)^{\gamma-1} \log p - \frac{(1 - p)^\gamma}{p}. \quad (15)$$

For an easy positive ($p \rightarrow 1$) we Taylor-expand $\log p = -(1 - p) - \frac{1}{2}(1 - p)^2 + O((1 - p)^3)$ and keep the leading term:

$$\left| \frac{\partial L_{\text{FL}}}{\partial p} \right| \sim (1 - p)^\gamma \left(\gamma + \frac{1}{p} \right) \xrightarrow{p \rightarrow 1} (1 - p)^\gamma (1 + \gamma). \quad (16)$$

For cross-entropy $L_{\text{CE}} = -\log p$ the derivative magnitude is $1/p \rightarrow 1$ at the same easy positive. The ratio of effective sample weights is

$$\frac{|\partial L_{\text{FL}}/\partial p|}{|\partial L_{\text{CE}}/\partial p|} \sim (1 - p)^\gamma (\gamma p + 1). \quad (17)$$

With our $\gamma = 2$ and an already-confident positive at $p = 0.9$, the ratio is $0.01 \cdot 2.8 = 0.028$, i.e. a $\sim 36\times$ down-weighting. The same calculation for an easy background sample ($y = 0$, $p_t = 1 - p \rightarrow 1$) is symmetric. This is the precise sense in which focal loss “down-weights easy examples”: the suppression is a power law of order γ in the per-example margin $(1 - p_t)$, with no additional hyperparameters.

11.3 GIoU as a Differentiable Surrogate for IoU

The IoU between two axis-aligned boxes A, B is $\text{IoU}(A, B) = |A \cap B|/|A \cup B|$. When $A \cap B = \emptyset$ the IoU vanishes identically on a non-empty open set of box parameters, so its gradient with respect to the predicted box coordinates is zero almost everywhere on that set — pure IoU is therefore an unusable regression loss until the predicted box already overlaps the ground truth.

Generalised IoU repairs this by introducing the smallest enclosing axis-aligned box $C(A, B)$ and adding a penalty for the area of C not occupied by $A \cup B$:

$$\text{GIoU}(A, B) = \text{IoU}(A, B) - \frac{|C(A, B) \setminus (A \cup B)|}{|C(A, B)|} \in [-1, 1]. \quad (18)$$

When A and B are disjoint, the IoU term is identically zero but the penalty $|C \setminus (A \cup B)|/|C|$ is strictly positive and strictly decreases as A approaches B (one can show $\text{GIoU} = -1$ when A and B are infinitely far apart along a common axis, and $\text{GIoU} \rightarrow 0^-$ as their gap shrinks to zero). The gradient $\partial L_{\text{GIoU}}/\partial(l, t, r, b)$ with respect to the predicted edge distances is therefore non-zero at every spatial configuration, providing a usable training signal even on the hard “no overlap” regime. The torchvision v2 reference implementations of RetinaNet and Faster R-CNN, as well as the FCOS regression head, all use $L_{\text{GIoU}} = 1 - \text{GIoU}$ in place of smooth- L_1 for this reason; the change is a strict mathematical improvement, with no extra hyperparameters.

11.4 Cosine Annealing with Warm-Up

The cosine schedule is

$$\eta(t) = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})(1 + \cos(\pi t/T)), \quad t \in [0, T]. \quad (19)$$

Its time derivative is

$$\eta'(t) = -\frac{\pi(\eta_{\max} - \eta_{\min})}{2T} \sin(\pi t/T), \quad (20)$$

which satisfies $\eta'(0) = \eta'(T) = 0$ and attains its peak magnitude $\pi(\eta_{\max} - \eta_{\min})/(2T)$ at $t = T/2$. Two properties follow.

Smooth start. $\eta'(0) = 0$ means the learning rate is locally constant at $\eta_{\max}$ during the first iterations, so the optimiser dynamics do not undergo an abrupt change at $t = 0$. The step schedule with milestone t_1 lacks this property: η' is a Dirac delta at t_1 and the optimiser experiences a discontinuous transition that can destabilise the EMA filter (whose effective window straddles the discontinuity for $\sim T_{\text{eff}}$ steps either side).

Smooth end. $\eta'(T) = 0$ means the final iterations are taken at a learning rate that is locally constant and small, which lets the optimiser settle into a flat region of the loss surface without the discontinuous shock of a final step decay. Combined with EMA this gives the visibly smoother end-of-training behaviour of the Adam-cosine curves in Figure 2 compared to the FCOS paper-SGD curve, whose two step-decay events at epochs 24 and 32 produce the visible jumps in the loss curves of Figure 1.

Warm-up. We prepend a linear warm-up $\eta(t) = (t/T_w) \eta_{\max}$ for $t < T_w = 2$ epochs because the Adam exponential moving averages $\hat{m}_t = (1 - \beta_1) \sum_k \beta_1^{t-k} g_k$ and $\hat{v}_t = (1 - \beta_2) \sum_k \beta_2^{t-k} g_k^2$ are biased toward zero in the first few steps, so the effective step size $\eta_t \hat{m}_t / \sqrt{\hat{v}_t + \epsilon}$ becomes large when $\hat{v}_t$ is still small. The warm-up keeps η_t small while $\hat{m}_t, \hat{v}_t$ stabilise, protecting the pretrained ResNet-50 features from a destructive early update.

11.5 Bootstrap Confidence Intervals: Monte Carlo Convergence

The percentile bootstrap CI is the empirical $[\alpha/2, 1 - \alpha/2]$ quantile of the resampled statistic $\hat{\theta}_b^*$ over B bootstrap resamples. By the central limit theorem applied to the resample distribution, the Monte Carlo standard error of any single quantile estimate decays as $O(1/\sqrt{B})$; in the standard Edgeworth analysis, the standard error of the p^{th} quantile is approximately

$$\text{SE}_p^{\text{MC}} \approx \frac{1}{f(q_p)} \sqrt{\frac{p(1-p)}{B}}, \quad (21)$$

where f is the density of the resampled statistic at the true quantile q_p . For our paired AP@0.5 differences with $\sigma \approx 1$ AP@0.5 point and a roughly Gaussian resample distribution, $1/f(q_{0.975}) \approx \sigma \sqrt{2\pi} \exp(z_{0.975}^2/2) \approx 6.3\sigma$, so at $B = 200$:

$$\text{SE}_{0.975}^{\text{MC}} \approx 6.3 \cdot 1 \cdot \sqrt{0.025 \cdot 0.975/200} \approx 0.07 \text{ AP points}. \quad (22)$$

The Monte Carlo error on either endpoint of our paired CIs is therefore well below the ~ 1 AP qualitative resolution we need, and re-running at $B = 1000$ would tighten the endpoints by a factor of $\sqrt{5} \approx 2.2\times$ to ~ 0.03 AP — a refinement that would not change a single one of the qualitative conclusions in Table 4. The CI width itself (~ 4 AP points on the paired differences) is set by the patient-resampling variance, *not* by B , and the only way to tighten it is to enlarge the validation set or run additional training seeds.

12 Discussion

12.1 Anchor-Free vs. Anchor-Based: A Revised View

The reference paper [1] argues that the anchor-free FCOS detector outperforms anchor-based alternatives on RSNA. Our experiments revise this claim in the opposite direction. On the full RSNA validation split, our anchor-free FCOS reaches 35.3% (Adam) and 39.8% (paper SGD); RetinaNet sits at 41.2%, Faster R-CNN at 42.9%. All four numbers exceed the paper’s reported FCOS score (28.5%). Under a shared Adam recipe the anchor-based detectors decisively win; under the paper’s SGD recipe FCOS narrows the gap to RetinaNet but does *not* demonstrably equalise the paradigms — we only ran the SGD ablation on the FCOS side, so the comparison “FCOS paper SGD vs. RetinaNet Adam” is intrinsically mixed. We emphasise this asymmetry because the SGD ablation is the addition that rescues the FCOS narrative; the honest test of paradigm-vs-optimiser would also retrain RetinaNet and Faster R-CNN under SGD, which is left for future work. Three further observations contextualise the result.

(i) Training regime matters more than paradigm. We trained with modern recipes that the paper does not mention: BF16 mixed precision, 2-epoch warm-up with cosine decay (or SGD step-decay for the paper-recipe ablation), EMA weights at $\mu = 0.999$, medical-specific augmentations, weighted sampling of positives, TTA, and Soft-NMS. Anchor-based detectors, with their stronger spatial priors, *benefit more* from these upgrades because the optimisation problem is easier to begin with. FCOS’s centre-ness branch takes longer to converge (Figure 2: FCOS-Adam crosses the paper’s 28.5% at epoch 12, while the same threshold is reached by both anchor-based detectors before epoch 8), so the same epoch budget yields a smaller gain for FCOS under Adam.

(ii) Convergence speed differs systematically. The four trajectories of Figure 2 read off cleanly. RetinaNet peaks at epoch 12 (37.3% pre-TTA) and then slowly drifts down to 34.5% at epoch 40; Faster R-CNN peaks identically early (epoch 12, 42.2%) and degrades more visibly to 36.7%. FCOS-Adam, by contrast, is still rising at the budget cap: its best validation AP is also its last (35.7% at epoch 40). The paper-SGD ablation gives FCOS the same 40-epoch budget but reshapes the trajectory: AP rises steadily to 39.9% at epoch 24 (right at the first SGD step-decay milestone) and stays flat thereafter. Two things follow: anchor-based detectors saturate within ~ 12 epochs and an even longer budget would not help them, whereas FCOS-Adam has visible headroom that a longer schedule or a stronger optimiser (as in the paper-SGD ablation) can convert into real AP.

(iii) Paper baselines were likely under-trained. The paper’s reported RetinaNet and Faster R-CNN scores (≈ 26 and 24% AP@0.5 respectively) are lower than our numbers by more than 15 points. This is a common pattern in publications of a new method: baselines are run with the paper’s own (often weaker) recipe rather than with each baseline’s best-known recipe. Our observation is *consistent* with [1]’s FCOS number while also showing that their baselines can be improved substantially.

12.2 What Did the Paper Get Right?

Despite the paper’s baselines being under-trained, its core methodological contribution is still valid. The advantages it attributes to FCOS are:

- *No anchor hyperparameters.* We confirm this. Our FCOS run uses the torchvision default with no task-specific tuning, while RetinaNet ideally should have anchors retuned for the RSNA size/aspect-ratio distribution (we used the COCO defaults); if anything this gives FCOS a stronger case in production settings.

- *Flexible localisation for variable-size lesions.* We list this as a paradigm-level argument, but we do *not* consider AR_L a clean confirmation of it. FCOS-Adam reaches $AR_L = 91.8\%$ and FCOS paper-SGD 93.8% , both above Faster R-CNN (87.0%) and well above RetinaNet (83.5%); however $\sim 99\%$ of RSNA boxes are in the COCO “large” bucket, so AR_L effectively measures overall recall at 10 detections, and the same FCOS recall-greediness that produces a high AR_L produces the low fixed-threshold precision of Table 2 (25.2% at $\tau = 0.3$). High AR_L here is therefore better read as a calibration / threshold artefact than as a localisation-quality win; the qualitative over-extension on row 2 of Figure 7 is the opposite signal. A fair test of the “flexible localisation” claim would need higher-IoU AP ($AP@0.75$) or a recall-controlled IoU, neither of which we ran.
- *Simpler architecture and loss.* We confirm this qualitatively: the FCOS head has one classification output plus four regression scalars plus a center-ness scalar – fewer hyperparameters than RetinaNet’s $9 \times A$ anchors and Faster R-CNN’s two-stage ROI head.

In short, the paper’s paradigm argument is an engineering-simplicity argument; our results simply show that when baselines are trained as well as the proposed method, the paradigm choice is not the dominant accuracy lever.

12.3 Clinical Interpretation of the Numbers

For a hypothetical clinical deployment, the metric that matters is not $AP@0.5$ but patient-level decision quality. The ensemble’s F1 of 63.0% (precision 56.1% , recall 72.0%) at a prevalence of $\sim 22\%$ translates to a positive predictive value of 56% and a sensitivity of 72% . Faster R-CNN used alone in its optimal operating point ($\tau^* = 0.824$) would deliver substantially better precision but at a cost of sensitivity; a radiologist-in-the-loop workflow would probably favour the ensemble’s higher sensitivity for triage, and reserve Faster R-CNN for a confirmatory second read. No detector we tested reaches the quality required for autonomous decision-making on this dataset (all ROC AUC values are below the $\approx 95\%$ threshold that is typically cited for diagnostic utility), a finding consistent with the broader literature on chest X-ray deep learning.

12.4 Cost and Reproducibility

The final four-model parallel training run completed in about 2.5 wall-clock hours on four H100 80 GB SXM GPUs, for a GPU-rental cost of approximately \$30 on RunPod’s community cloud. Adding the cached-prediction inference pass and the post-hoc analyses, the entire pipeline from raw DICOM to compiled PDFs runs end-to-end in under three hours of GPU time. The public codebase (Section 14) ships a deterministic patient-level 80/20 split (seed 42), incremental `history.json` writes after every validation (so an interrupted run is recoverable), the cached predictions and `all_metrics_v2.json` used to draw every figure in Section 10, and a single-command pipeline (`main.py -mode full`) for the training side. The split and the post-hoc analyses are bit-deterministic; the training run itself is *not* — TF32 + BF16 + asynchronous dataloader workers + nondeterministic CUDA atomics together break exact bit-level reproducibility, and a re-run of `main.py` on the same hardware will land within ~ 1 point of the cached numbers but not on top of them. Our statistical claims therefore rest on the cached predictions distributed with the report, not on the assumption that the run can be regenerated bit-for-bit.

13 Limitations and Future Work

13.1 Limitations

Fixed backbone and input resolution. All detectors share a ResNet-50 backbone at 512×512 input resolution, matching Wu et al. A larger backbone (ResNet-101, Swin) or a 1024×1024 input would likely improve every detector at the cost of memory and compute. We retained the paper’s setting so that the paradigm comparison is not confounded by an unrelated backbone change.

Single training seed and no held-out test set. Every reported number comes from a single training run per detector with a fixed seed of 42, evaluated on the same patient-level 20% validation split that was used for early-stopping / checkpoint selection, threshold fitting, the 5-fold-CV learnt aggregator, and the bootstrap CIs. Section 10 mitigates this in two limited ways: (i) the patient-level bootstrap quantifies how stable each detector’s AP@0.5 is under *patient resampling*, and (ii) the three-way split refits the Youden threshold on a calibration half so that the F1 number is not over-fit. Two important sources of variance remain uncovered: *run-to-run variance* (initialisation, augmentation RNG, optimiser stochasticity), which is routinely $\pm 1\text{--}2$ AP for modern detectors and which a single seed cannot estimate; and the *selection-on-val* bias from picking the best EMA checkpoint of each detector on the same set the headline AP is read on (with validation every 4 epochs the max is over ~ 10 noisy estimates, so the reported AP@0.5 of every detector in Table 1 is upward-biased by an amount we do not quantify). The honest reading is therefore that AP@0.5 differences smaller than ~ 2 points should not be treated as resolved, and the pairwise comparisons of Section 10 are reported without a Holm/Bonferroni correction for the four pairwise tests we run, so the $p = 0.080$ for Faster R-CNN vs. RetinaNet is best read as “not significant” rather than “marginally significant”. A full k -seed retrain, a held-out test set carved out before any model selection, and a Bonferroni correction across pairwise tests remain future work.

No per-site stratification or external dataset. The RSNA dataset was collected from 16 institutions [7] but the challenge data does not expose the institution, so we cannot stratify error by site. We also do not evaluate on an external chest-X-ray dataset (ChestX-ray14, CheXpert). Both are necessary to characterise site- and domain-shift generalisation for real deployment.

13.2 Future Work

Cross-model knowledge distillation. The F1-dominant ensemble can be distilled into a single model that is cheaper at inference and approximates the ensemble’s calibration. Mean Teacher-style self-distillation [24] is particularly well suited because EMA is already in our pipeline.

Stronger assignment strategies. ATSS [17] showed that much of the anchor-free vs. anchor-based gap disappears under a unified positive-sample assignment strategy. Replacing the default assigners in RetinaNet and FCOS with ATSS would test whether the remaining gap is truly about the detection head or about label assignment.

Transformer-based detectors. DETR [18] and its variants (Deformable DETR, DINO) produce set-based detections without NMS and have dominated COCO leaderboards since 2022. They would be a natural follow-up comparator on RSNA; their behaviour on small, structurally simple datasets like RSNA (compared to COCO) is less studied.

Clinical calibration study. A study that measures detector agreement with radiologist readings – rather than with the RSNA bounding boxes, which were themselves produced by a single rater per image – would establish the true clinical utility of these models. This is the standard bar for deployment and is currently out of reach of a student project because it requires IRB-approved chart review.

Parametric post-hoc calibration. Section 10 addresses score calibration with a learnt logistic-regression aggregator over per-patient features (*not* a per-score calibration map). A more classical parametric calibration — temperature scaling on the patient max-score, or Platt scaling — would be a useful complementary baseline: it would isolate the contribution of pure score recalibration from the multi-feature aggregation we already have. Both fit on the same calibration half we use for the Youden threshold and would not require additional training.

14 Conclusion

We reproduced the anchor-free pneumonia detection approach proposed by Wu et al. and systematically compared it against two anchor-based baselines (RetinaNet and Faster R-CNN) on the full RSNA Pneumonia Detection Challenge dataset. All models share the same ResNet-50 + FPN backbone, ensuring that performance differences stem from the detection paradigm.

Key findings:

1. All detectors exceed the paper’s reported FCOS AP@0.5 of 28.5%: our FCOS (Adam) reaches 35.3%, FCOS (paper SGD) 39.8%, RetinaNet 41.2%, and Faster R-CNN 42.9%. These point estimates carry two caveats that the patient-level bootstrap does not address: they are read at each detector’s best-validation EMA checkpoint, which is a max over ~ 10 noisy estimates and therefore upward-biased; and the bootstrap quantifies only test-sampling variance on a single training seed, so the 1–2 AP run-to-run variance typical of detection is *not* captured.
2. **Under a shared Adam recipe the anchor-based detectors outperform the anchor-free FCOS on AP@0.5; under the paper’s SGD recipe FCOS narrows the gap to RetinaNet, although our reading of the paired bootstrap (Section 10) does *not* support a strong “statistically indistinguishable” claim.** We have run the SGD ablation only for FCOS, so we cannot rule out that RetinaNet or Faster R-CNN would also gain under SGD; the SGD result narrows the FCOS-vs-RetinaNet gap rather than equalising the paradigms.
3. A Weighted Box Fusion ensemble of FCOS and RetinaNet reaches patient-level F1 63.0% at the fixed $\tau = 0.3$ threshold, +13.5 points above the best single model *at that arbitrary threshold*. This headline is largely an artefact of the calibration mismatch documented in Section 10: under a fair per-detector operating point (Youden threshold fitted on a calibration half, or the learnt aggregator), the ensemble’s advantage shrinks to ≤ 1 F1 point over the best single detector (RetinaNet under the learnt aggregator and Faster R-CNN under the held-out Youden threshold). The fixed-threshold $\tau = 0.3$ protocol is therefore the wrong place to make the ensemble case, even if it is the protocol the paper uses.
4. Threshold-free ROC AUC lies between 86.5% and 89.1% for all five entries, a much tighter range than the fixed-threshold F1 suggests. The learnt patient aggregator (Section 10) lifts every single-detector F1 by +10 to +22 points and brings all four detectors into a ~ 3 -point band, which is decisive evidence that the F1 gap was a calibration artefact of the naive “any box $> \tau$ ” rule rather than an intrinsic detection-quality gap.

The single-run nature of the experiments is partly mitigated by the Section 10 robustness checks — patient-level bootstrap CIs on AP@0.5, a three-way split protocol that fits the Youden threshold on a held-out calibration half, RSNA-percentile size buckets, FROC/CPM, reliability diagrams with ECE, and a learnt 5-fold-CV patient-level aggregator — but those bootstraps quantify only the variance of resampling validation patients on a single trained model, not the variance across training seeds. The remaining caveats — a single training seed per detector, no held-out test set, no multiple-comparison correction across the pairwise tests, no per-site stratification and no external-dataset validation — are stated in Section 13. End-to-end the pipeline runs in ~ 2.5 hours of wall-clock on 4xH100 for a GPU-rental cost of approximately \$30; that figure is a logistics note, not a contribution.

Availability

The code, the exact training script, and the compiled artefacts of this report are available at <https://github.com/ma155/pneumonia-detection-naml>. The RSNA dataset is freely available from the 2018 RSNA Pneumonia Detection Challenge at the Kaggle competition page kaggle.com/c/rsna-pneumonia-detection-challenge. We use the `torchvision` ResNet-50 ImageNet V1 weights and the `torchvision` reference implementations of FCOS, RetinaNet v2 and Faster R-CNN v2 as the starting points for the three detectors; no proprietary model weights are used.

References

- [1] J. Wu, L. Xia, X. Cheng, Y. Xu, and Z. Tian. Pneumonia detection based on RSNA dataset and anchor-free deep learning detector. *Scientific Reports*, 14:2581, 2024. [doi:10.1038/s41598-024-52156-7](https://doi.org/10.1038/s41598-024-52156-7).
- [2] Z. Tian, C. Shen, H. Chen, and T. He. FCOS: Fully convolutional one-stage object detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9627–9636, 2019.
- [3] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2999–3007, 2017.
- [4] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie. Feature pyramid networks for object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 936–944, 2017.
- [5] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.
- [6] S. Ren, K. He, R. Girshick, and J. Sun. Faster R-CNN: Towards real-time object detection with region proposal networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 91–99, 2015.
- [7] G. Shih, C. Wu, S. Halabi, et al. Augmenting the National Institutes of Health chest radiograph dataset with expert annotations of possible pneumonia. *Radiology: Artificial Intelligence*, 1(1):e180041, 2019.
- [8] R. Girshick, J. Donahue, T. Darrell, and J. Malik. Rich feature hierarchies for accurate object detection and semantic segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 580–587, 2014.
- [9] R. Girshick. Fast R-CNN. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1440–1448, 2015.
- [10] K. He, G. Gkioxari, P. Dollár, and R. Girshick. Mask R-CNN. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2961–2969, 2017.
- [11] Z. Cai and N. Vasconcelos. Cascade R-CNN: Delving into high quality object detection. In *CVPR*, pages 6154–6162, 2018.
- [12] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. Berg. SSD: Single shot multibox detector. In *ECCV*, pages 21–37, 2016.
- [13] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi. You only look once: Unified, real-time object detection. In *CVPR*, pages 779–788, 2016.
- [14] H. Law and J. Deng. CornerNet: Detecting objects as paired keypoints. In *ECCV*, 2018.
- [15] X. Zhou, D. Wang, and P. Krähenbühl. Objects as points. *arXiv:1904.07850*, 2019.
- [16] T. Kong, F. Sun, H. Liu, Y. Jiang, L. Li, and J. Shi. FoveaBox: Beyond anchor-based object detector. *IEEE Transactions on Image Processing*, 29:7389–7398, 2020.

- [17] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li. Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection. In *CVPR*, pages 9759–9768, 2020.
- [18] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko. End-to-end object detection with transformers. In *ECCV*, pages 213–229, 2020.
- [19] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, and R. M. Summers. ChestX-Ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localisation of common thorax diseases. In *CVPR*, pages 2097–2106, 2017.
- [20] N. Bodla, B. Singh, R. Chellappa, and L. S. Davis. Soft-NMS – improving object detection with one line of code. In *ICCV*, pages 5561–5569, 2017.
- [21] R. Solovyev, W. Wang, and T. Gabruseva. Weighted boxes fusion: Ensembling boxes from different object detection models. *Image and Vision Computing*, 107:104117, 2021.
- [22] I. Loshchilov and F. Hutter. SGDR: Stochastic gradient descent with warm restarts. In *ICLR*, 2017.
- [23] B. T. Polyak and A. B. Juditsky. Acceleration of stochastic approximation by averaging. *SIAM Journal on Control and Optimization*, 30(4):838–855, 1992.
- [24] A. Tarvainen and H. Valpola. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. In *NeurIPS*, pages 1195–1204, 2017.
- [25] P. Micikevicius, S. Narang, J. Alben, et al. Mixed precision training. In *ICLR*, 2018.
- [26] Y. Li, S. Xie, X. Chen, P. Dollár, K. He, and R. Girshick. Benchmarking detection transfer learning with vision transformers. *arXiv:2111.11429*, 2021.